

AI Capabilities, Computing Constraints, and the Economics of AI Deployment

July 24, 2026 – AI for Financial Services



Neil Thompson
PI and FutureTech Project
Director

AI
capabilities
improve



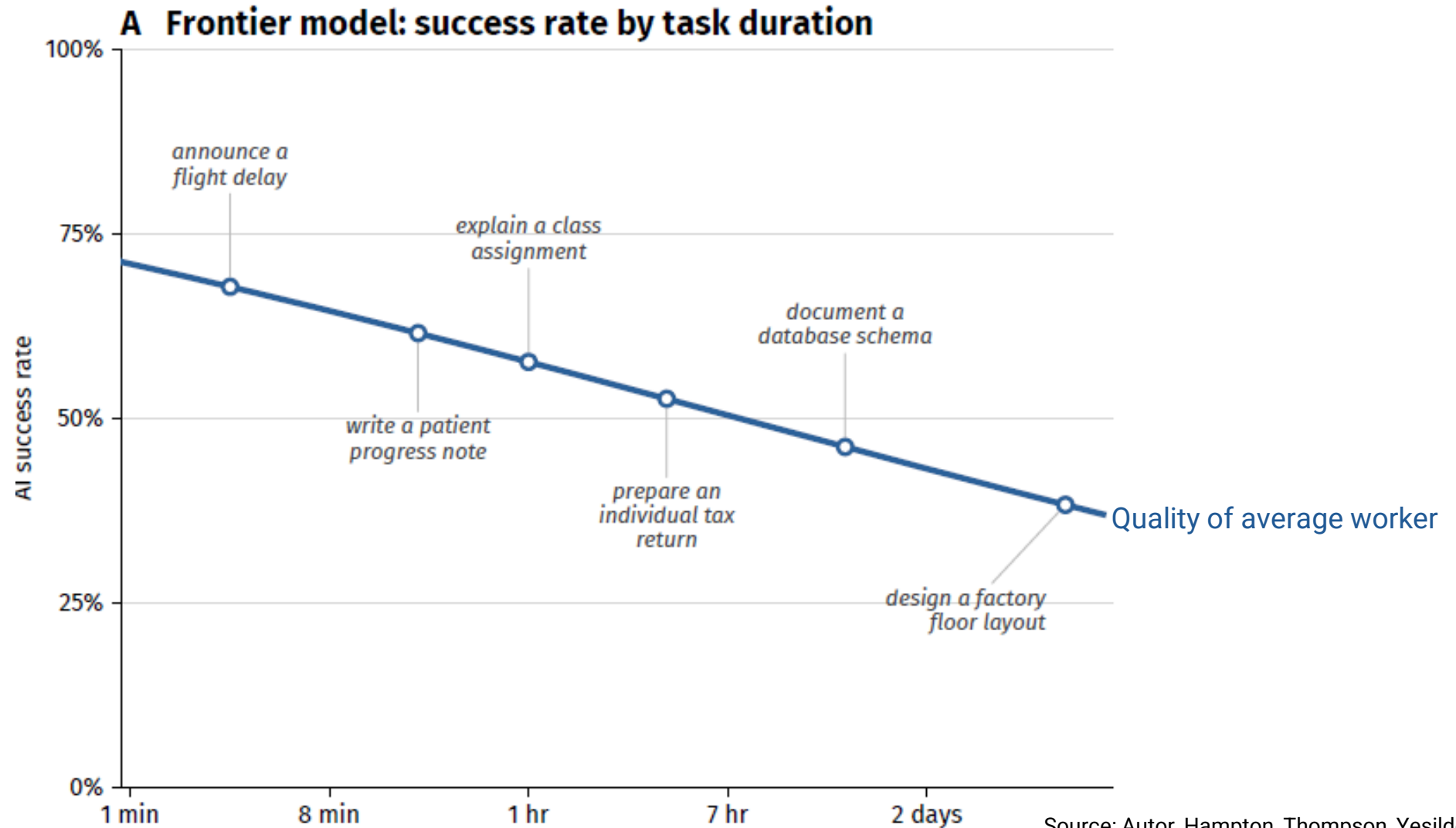
Firms
adopt



Firms and
workers
experience
effects

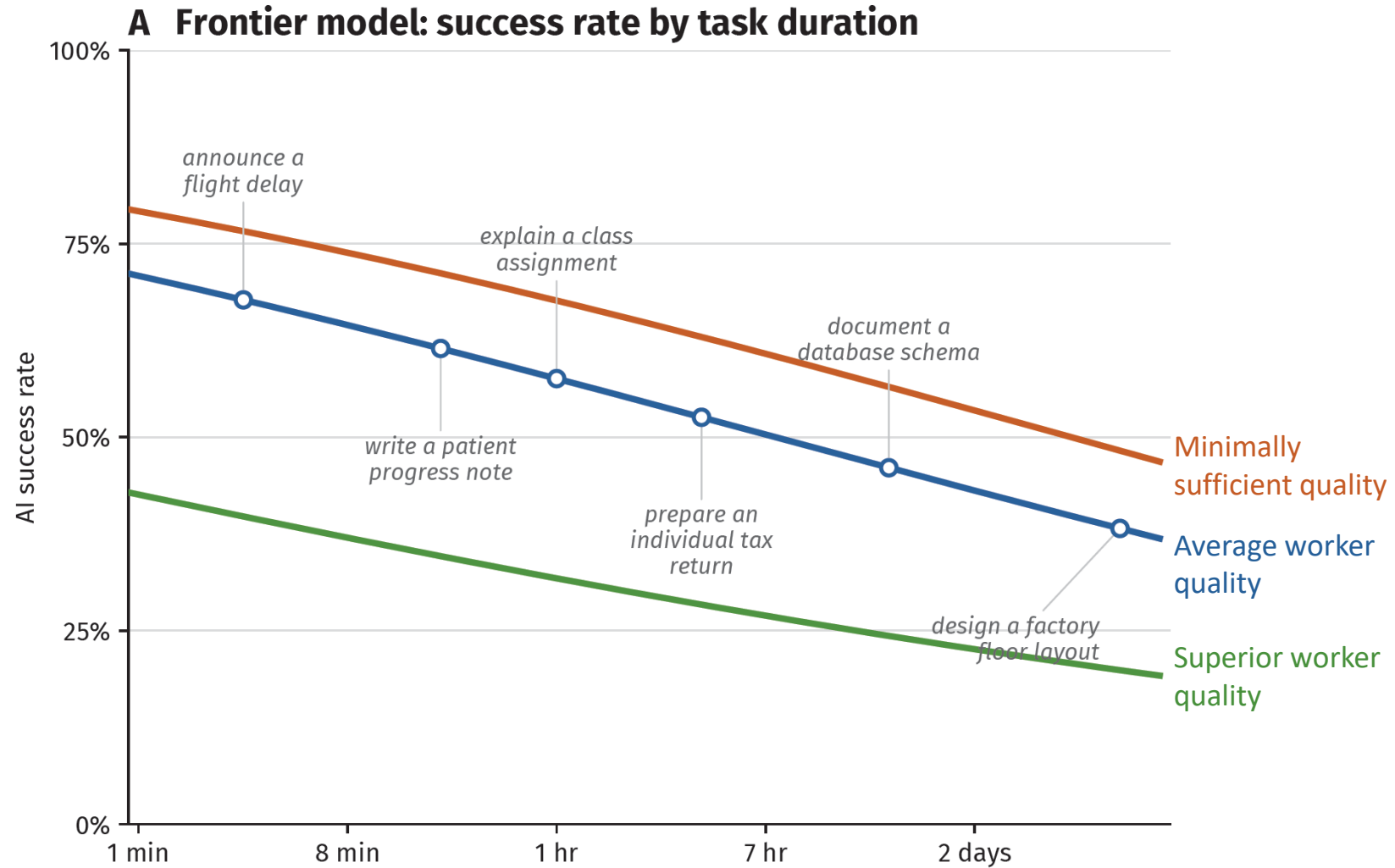
How good is AI?

AI Success Rates *if AI has all the needed information*



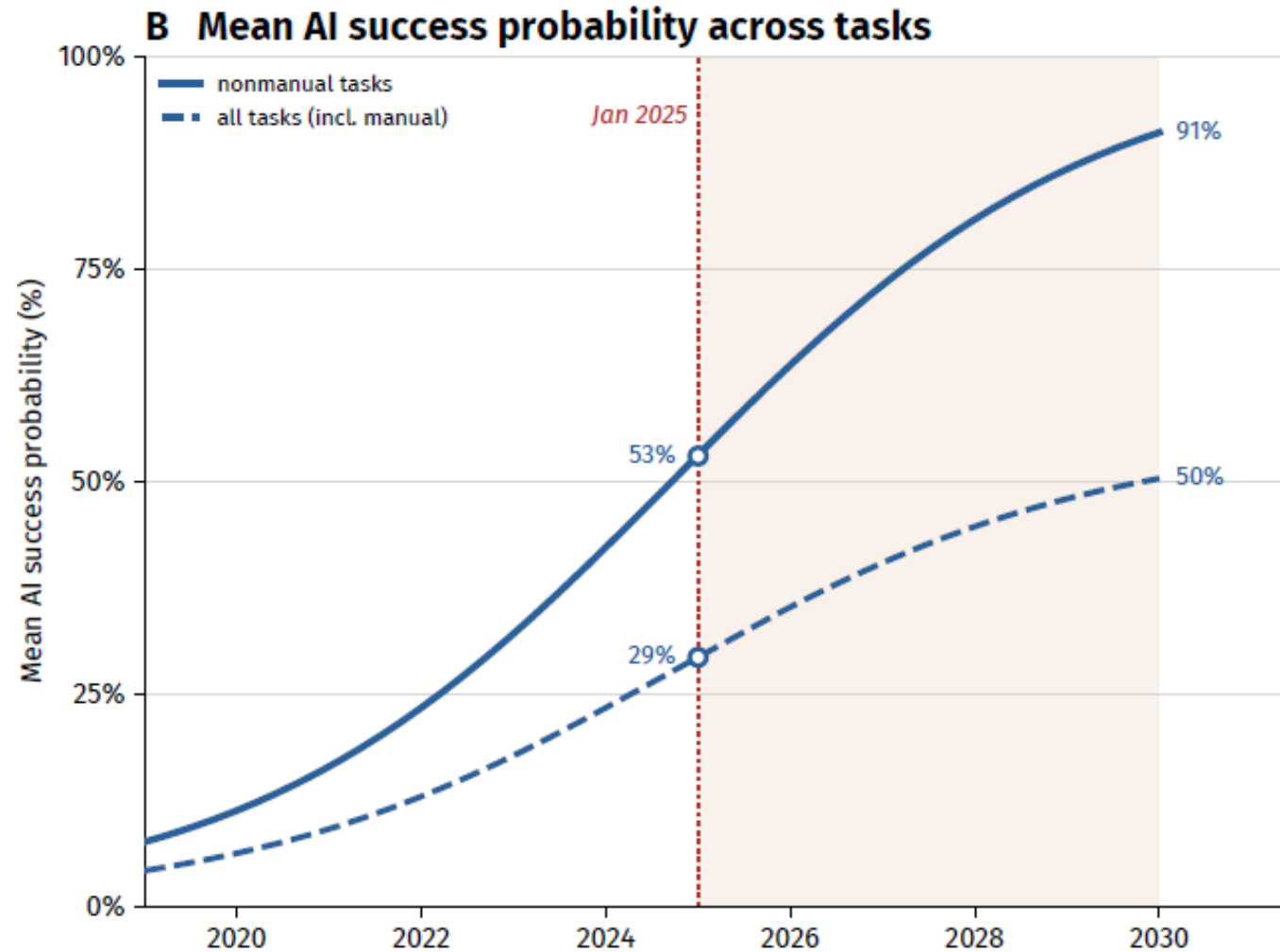
Source: Autor, Hampton, Thompson, Yesildere (2026)
based on Mertens et al. (2026)

AI Success Rates *if AI has all the needed information*



Source: Autor, Hampton, Thompson, Yesildere (2026)
based on Mertens et al. (2026)

Share of tasks where AI can match average human quality



Why is AI so good?

The Power of Scaling-up AI Neural Networks

A portrait photo of a kangaroo wearing an orange hoodie and blue sunglasses standing on the grass in front of the Sydney Opera House holding a sign on the chest that says Welcome Friends!



350 million parameters



750 million parameters

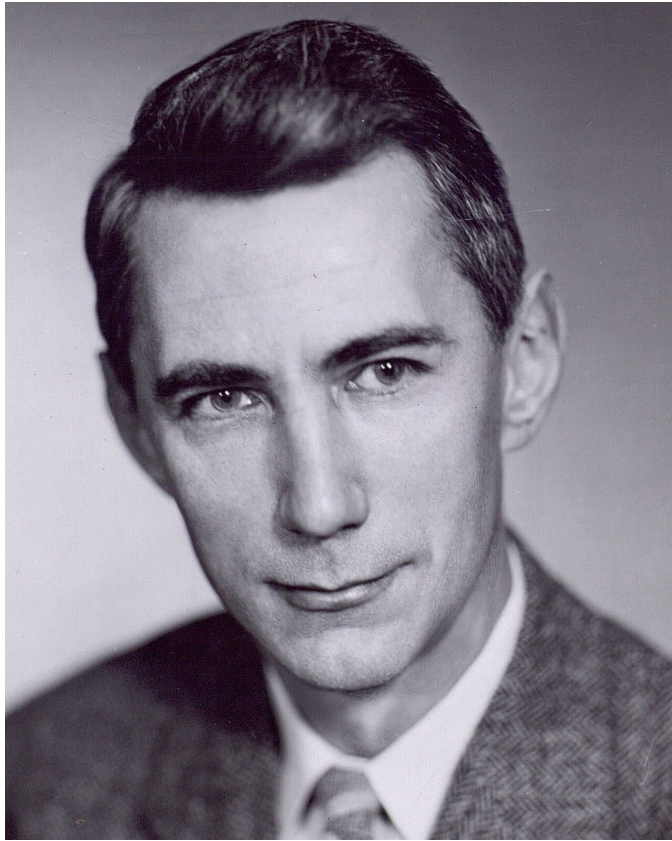


3 billion parameters



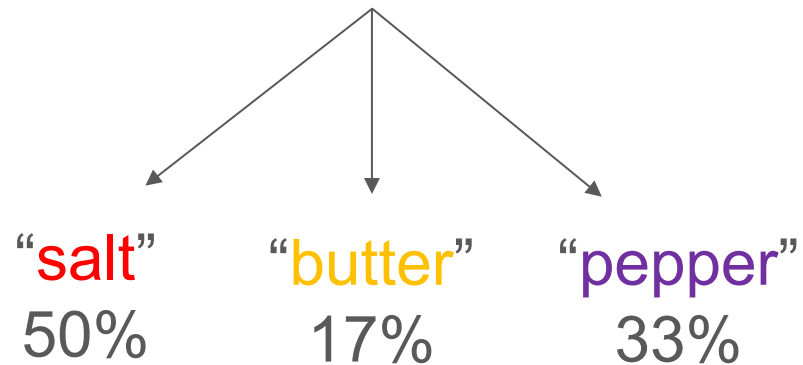
20 billion parameters

The ideas were already present the dawn of computing (1948)



Claude Shannon
“Father of Information Theory”
(1916 – 2001)

“Please pass the _____?”

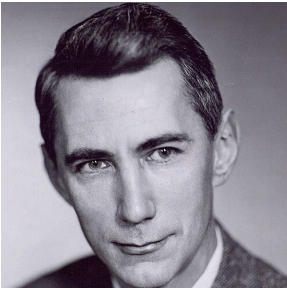


Please pass the salt.
Please pass the pepper.
Please pass the salt.
Please pass the pepper.
Please pass the butter.
Please pass the salt.

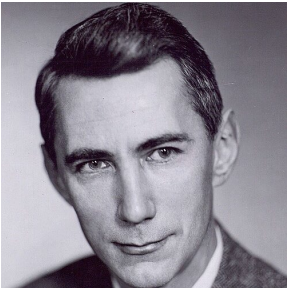
...

Diagram illustrating the equivalence of different AI models to a specific FLOPs value:

$$\boxed{\text{GPT 5.4}} \approx \boxed{\text{Gemini 3.1 Pro}} \approx \boxed{\text{Claude Opus 4.6}} \approx 10^{26} \text{ FLOPs}$$

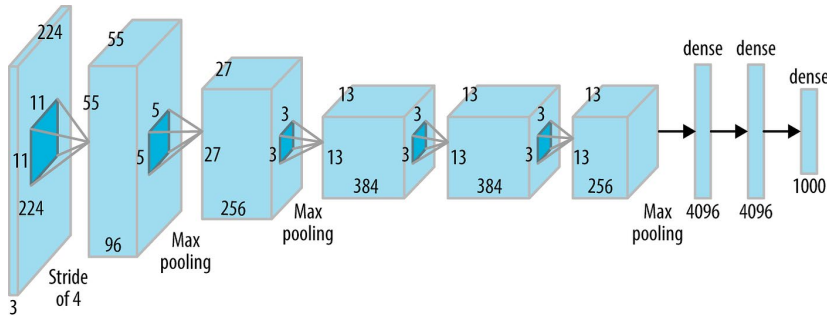
		Dollars
1950		\$9 billion trillion
2012		\$12 billion
2026		\$105 million



		Dollars	Dollars Spent		
1950		\$9 billion trillion	\$500		<u>Spend</u> 5×
2012		\$12 billion	\$2,500		<u>Efficiency</u> 720,000,000,000×
2026		\$105 million	\$105 million		<u>Spend</u> 42,000×
					<u>Efficiency</u> 110×

All values approximate

2012 AlexNet



6 days

2 GPUs

2016 Google Neural Machine Translation



6 days

96 GPUs

2023 GPT-4



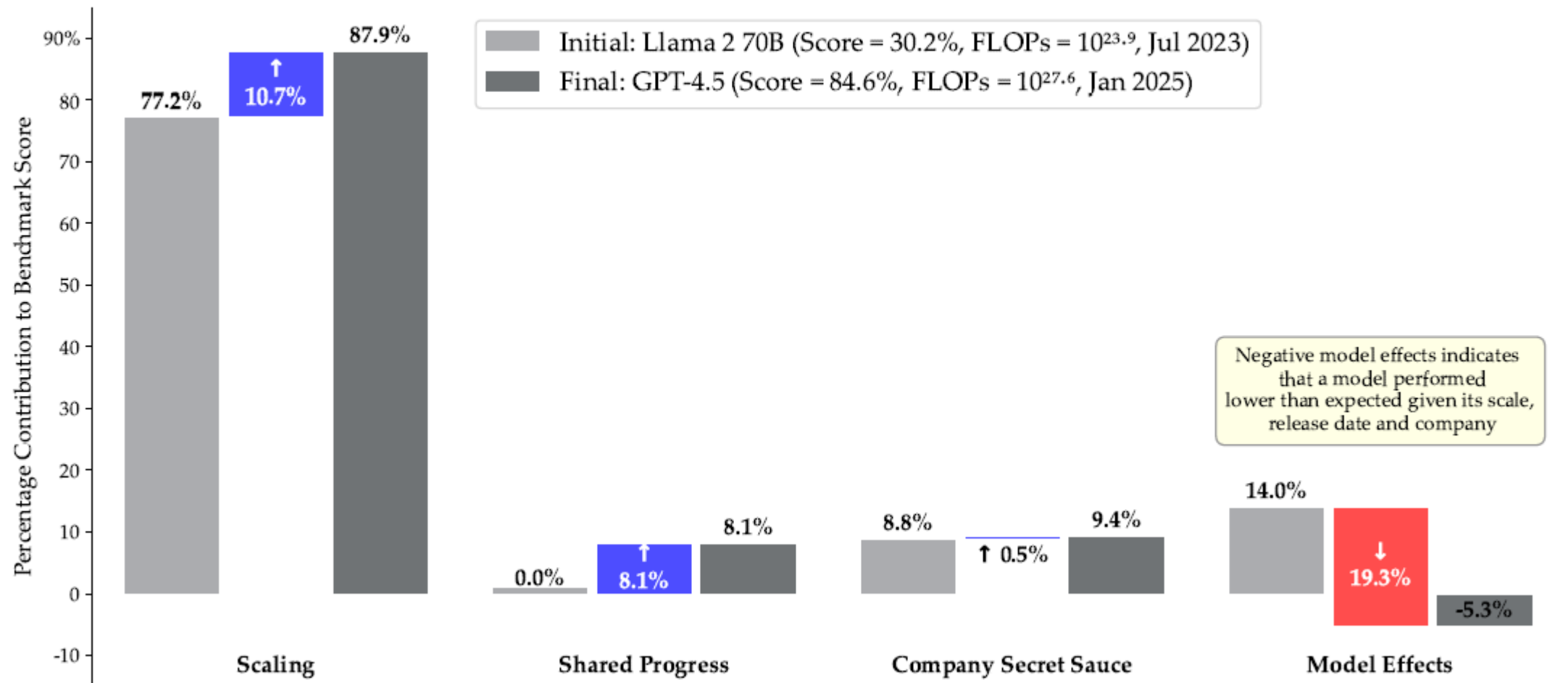
~100 days

25,000 GPUs

Contributors to AI improvement: Frontier Models

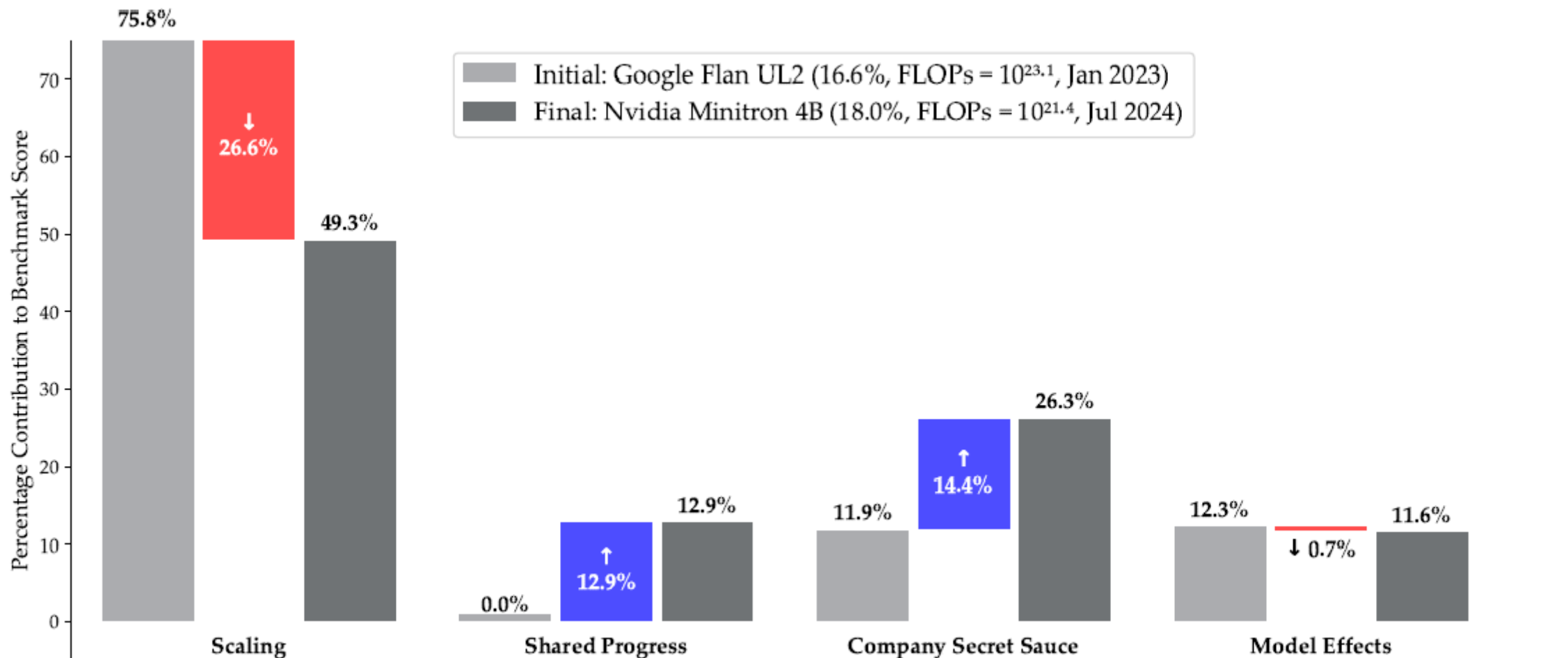
(Mertens, Fischl-Lanzoni, and Thompson 2026)

(a) Source of Benchmark Score Growth: How Top Performing Models Become Better

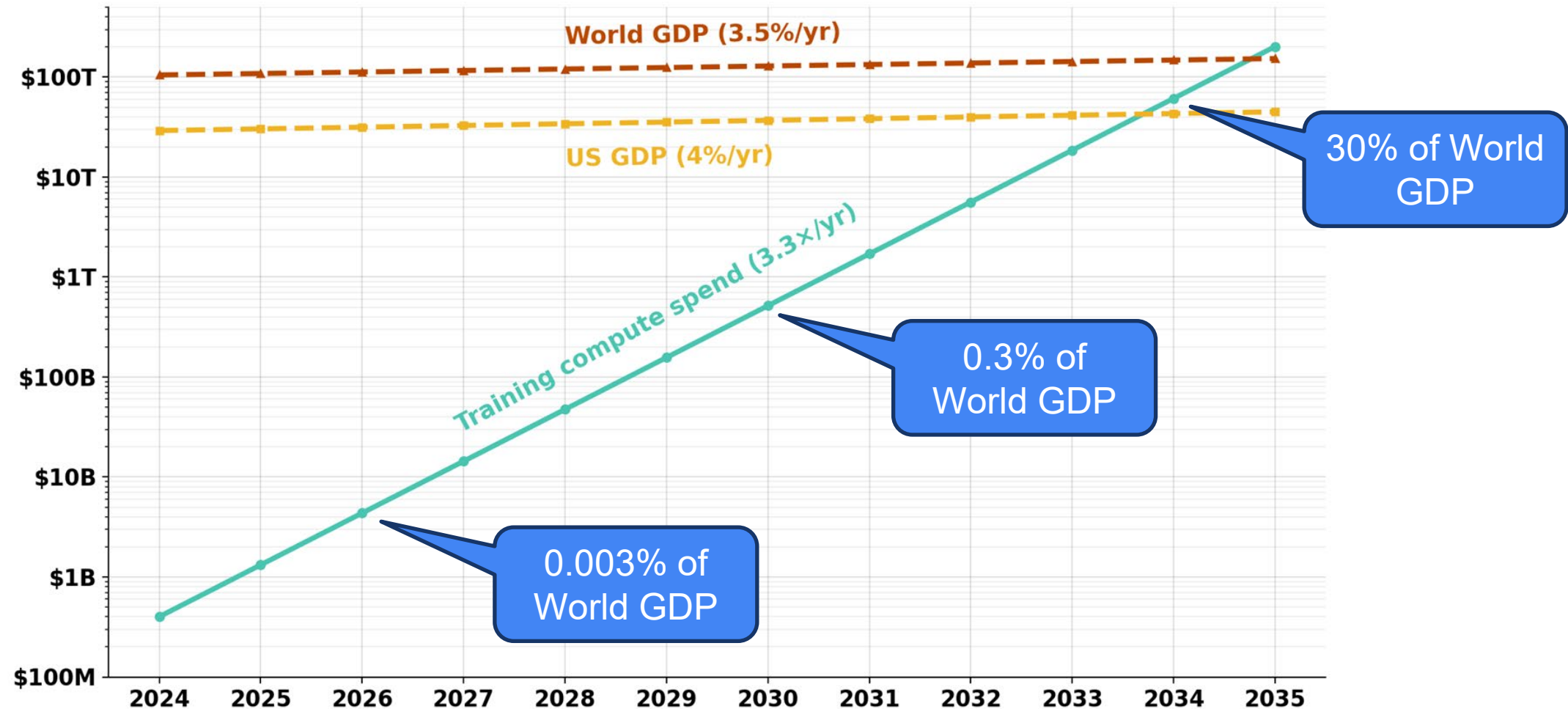


Contributors to AI improvement: Efficient Models

(Mertens, Fischl-Lanzoni, and Thompson 2026)

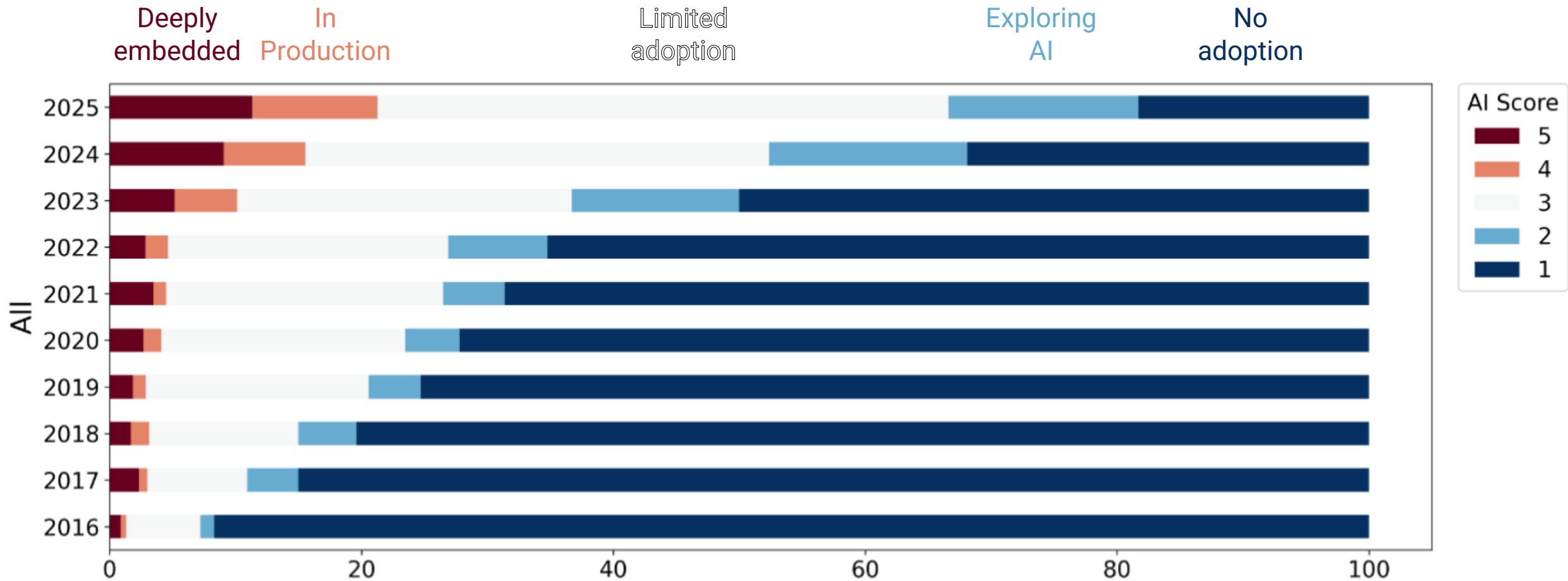


Scaling up costs will soon become prohibitively expensive



What will this mean for firms?

AI Adoption at the firm level (S&P 500)

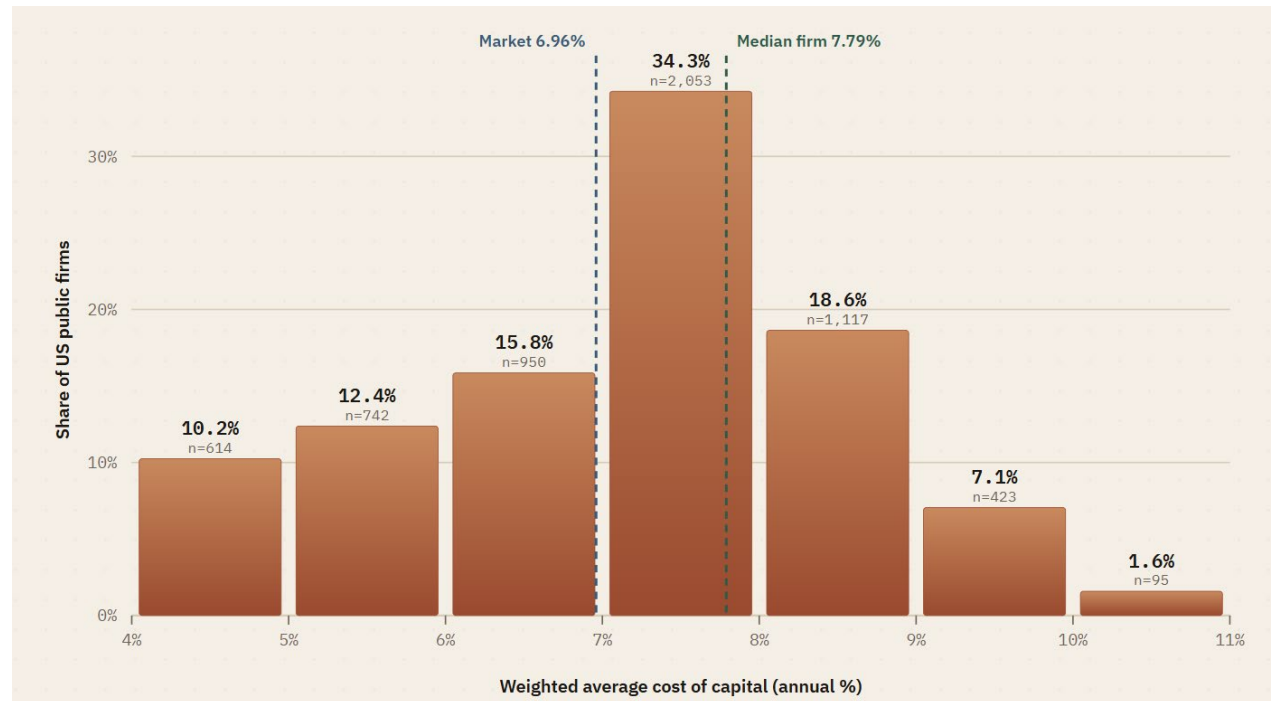




Average ROI for AI projects that have scaled?

7%

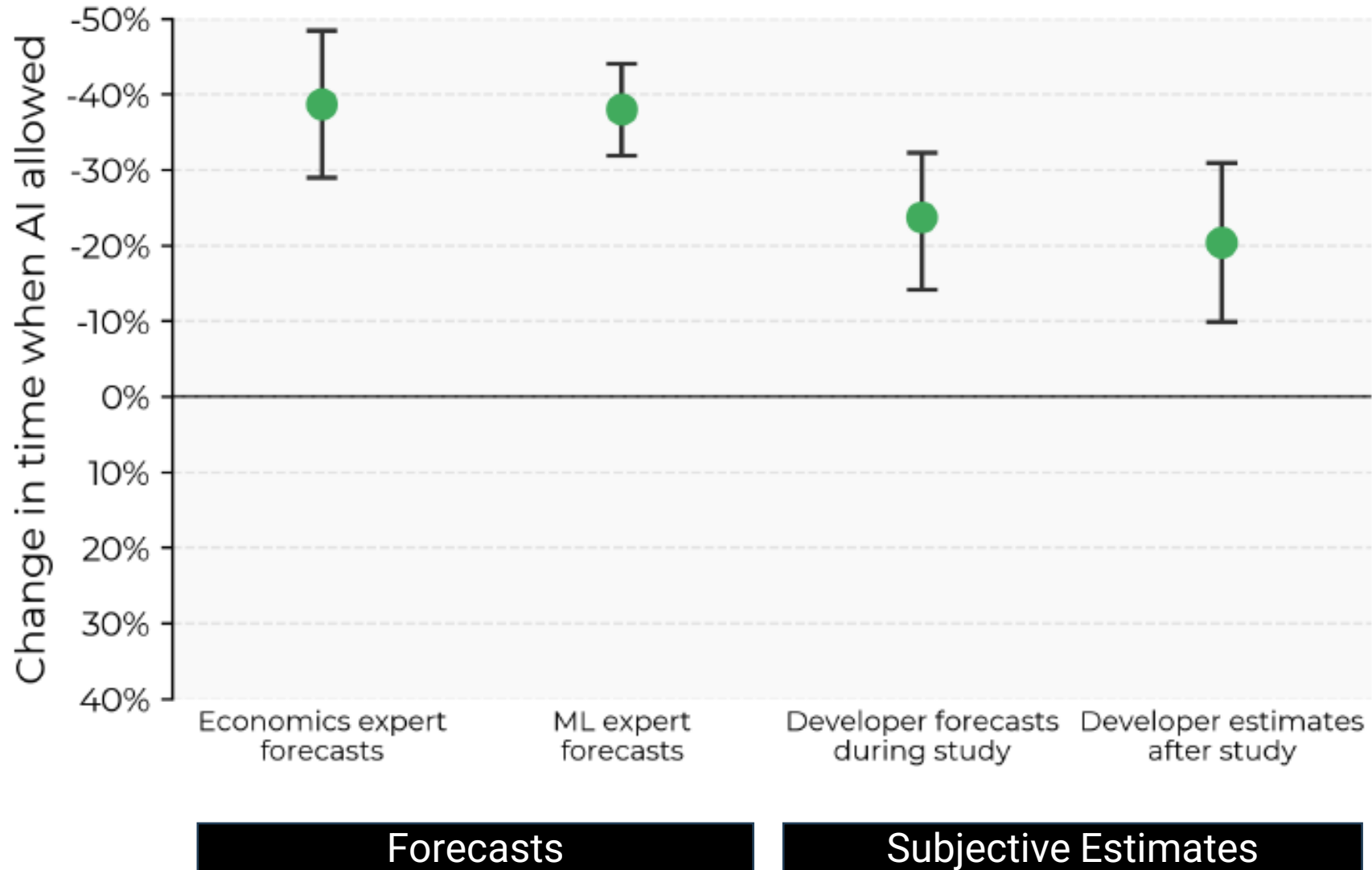
**U.S. Company
cost of capital**
(Based on Damodaran, 2026)



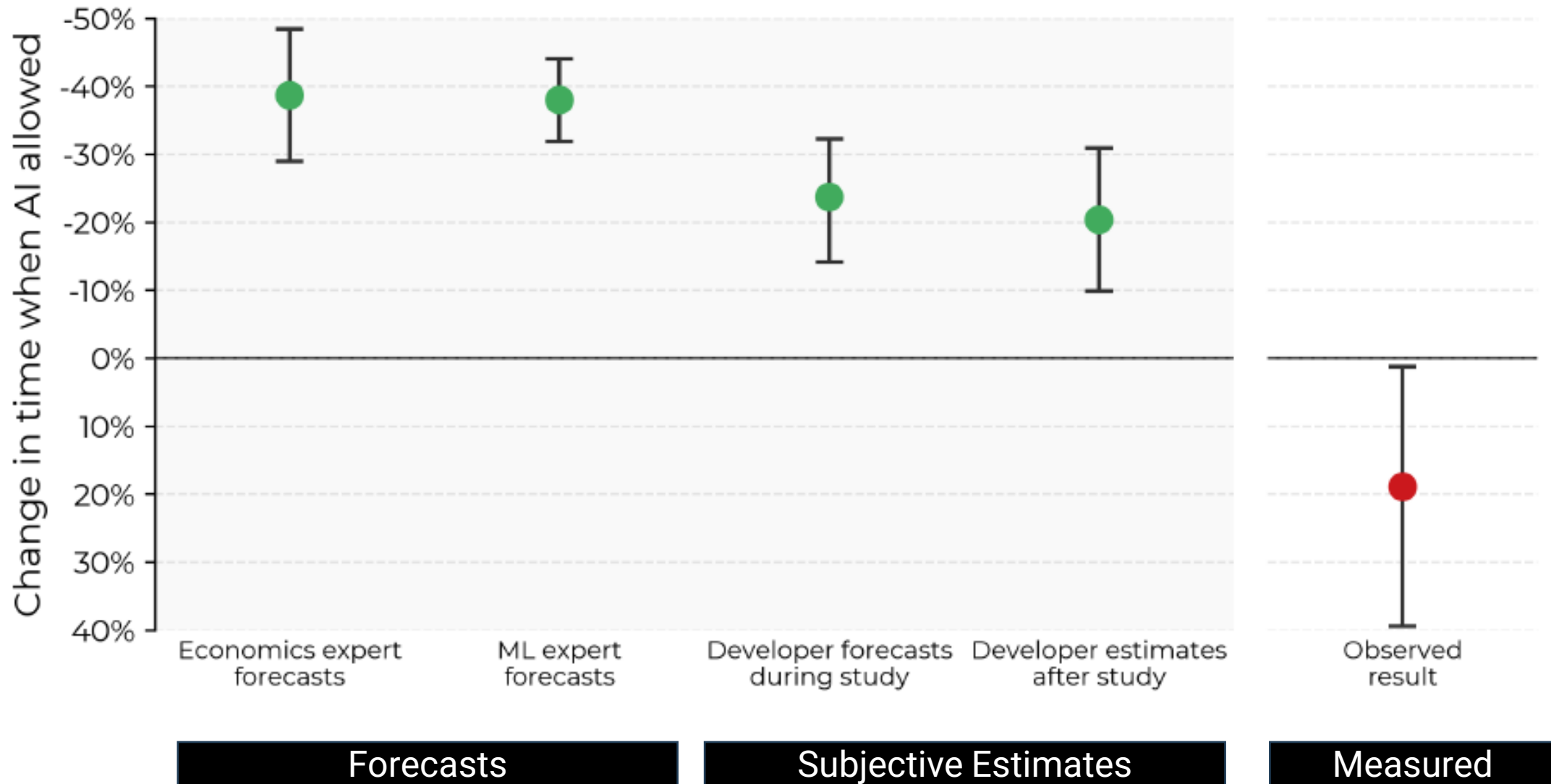
Challenge #1:
AI has good and bad effects on productivity

Randomized Control Trial on Software Developer Productivity

(246 tasks across 16 experienced developers – Becker et al., 2025)

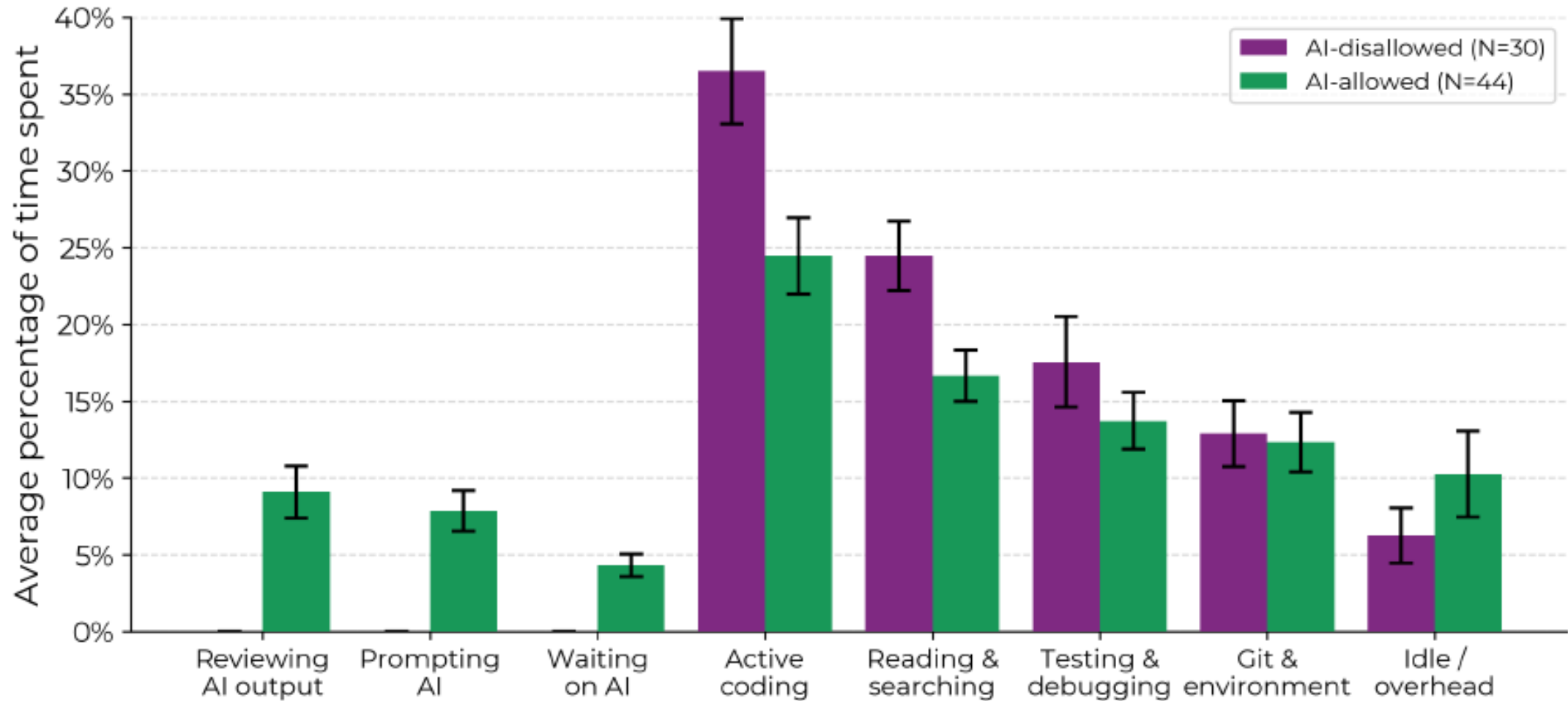


Randomized Control Trial on Software Developer Productivity (246 tasks across 16 experienced developers – Becker et al., 2025)



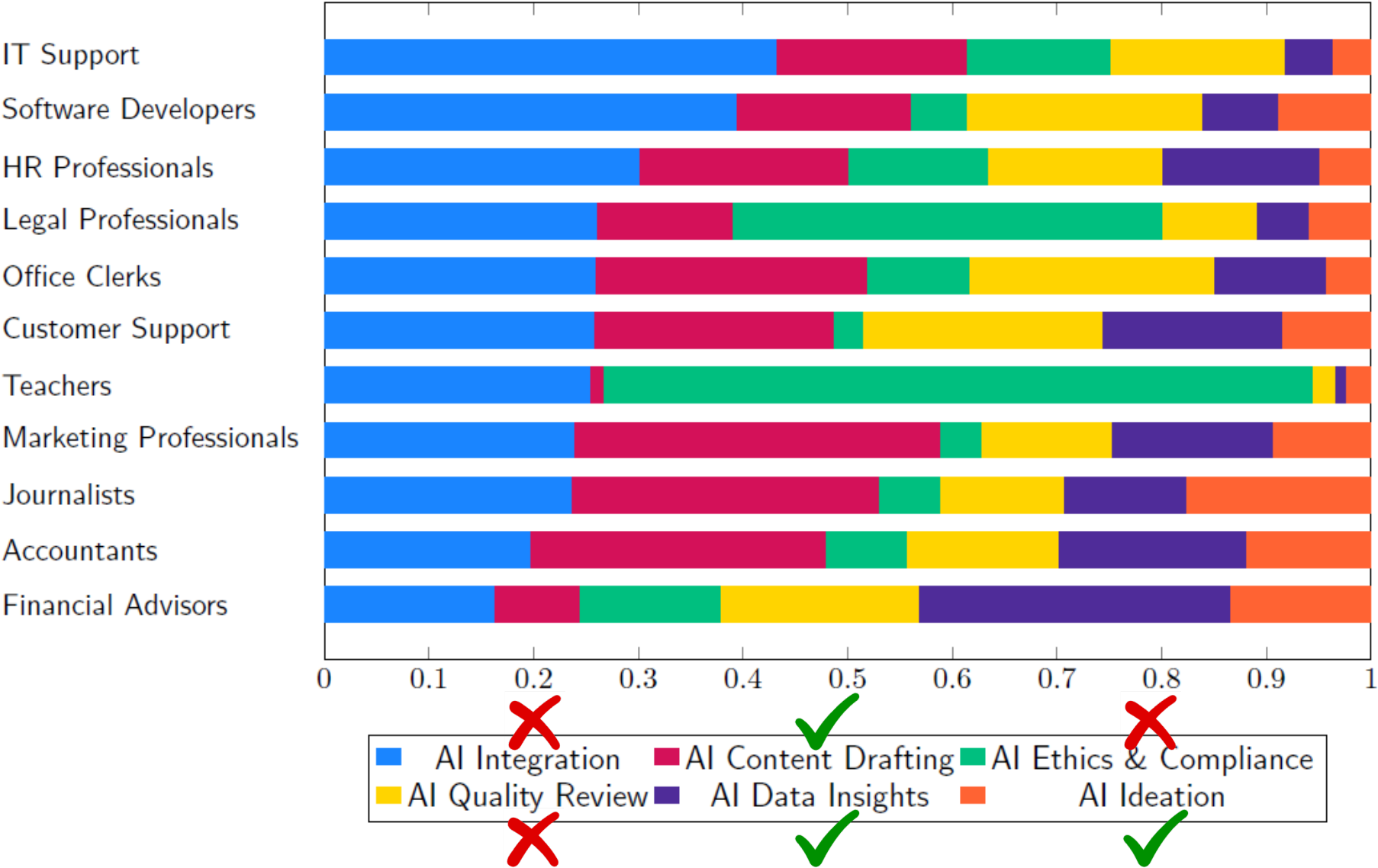
Randomized Control Trial on Software Developer Productivity

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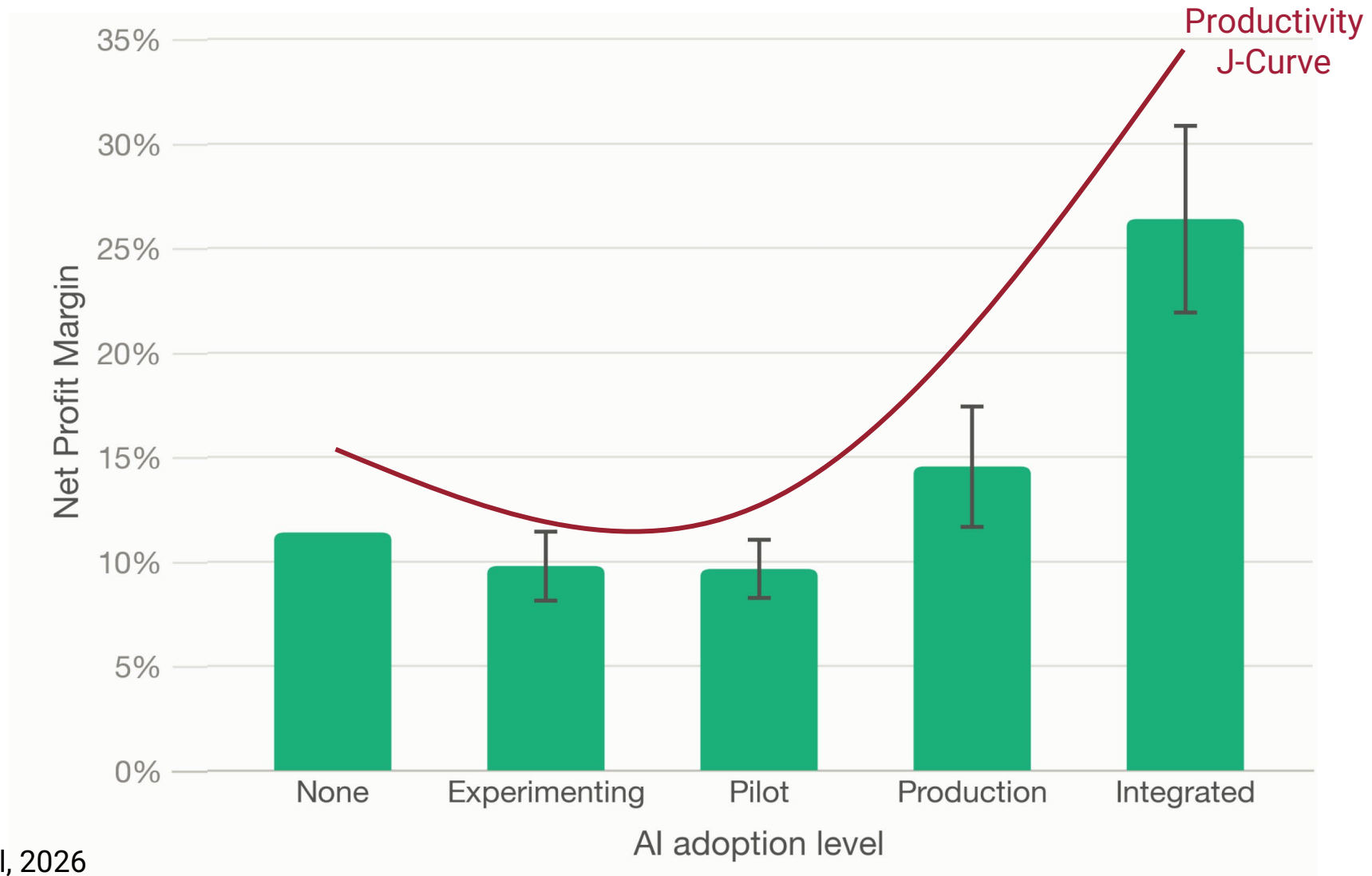


Composition of AI Tasks

(Humlum and Vestergaard, 2025)



AI adoption has counter-intuitive effects for Non-Tech firms



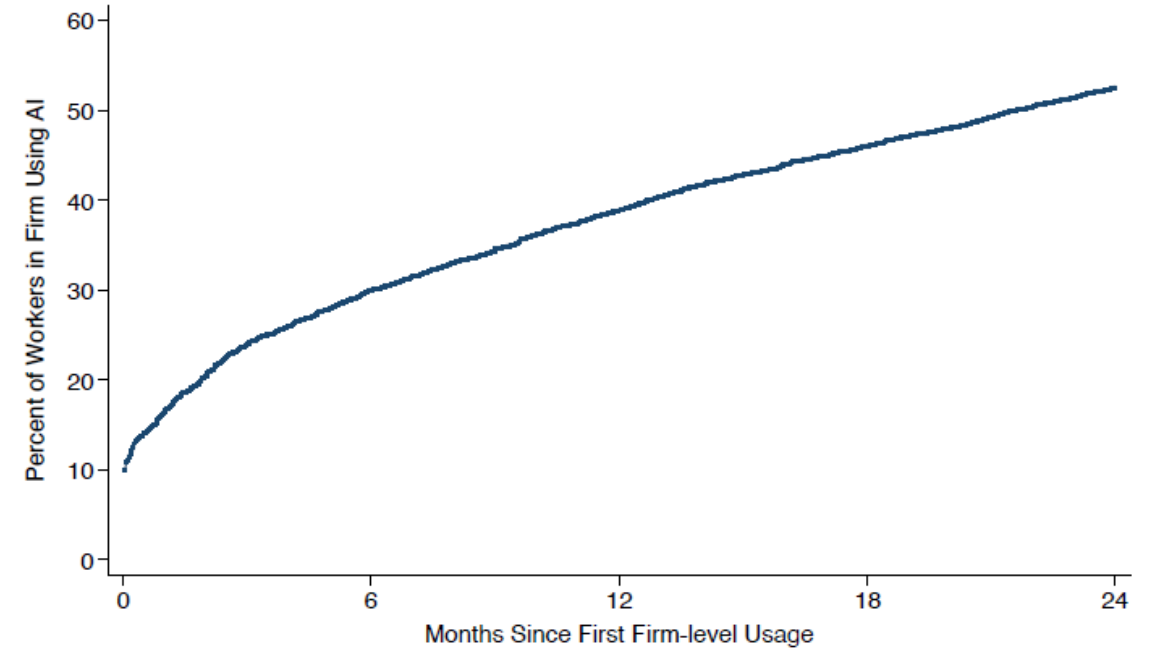
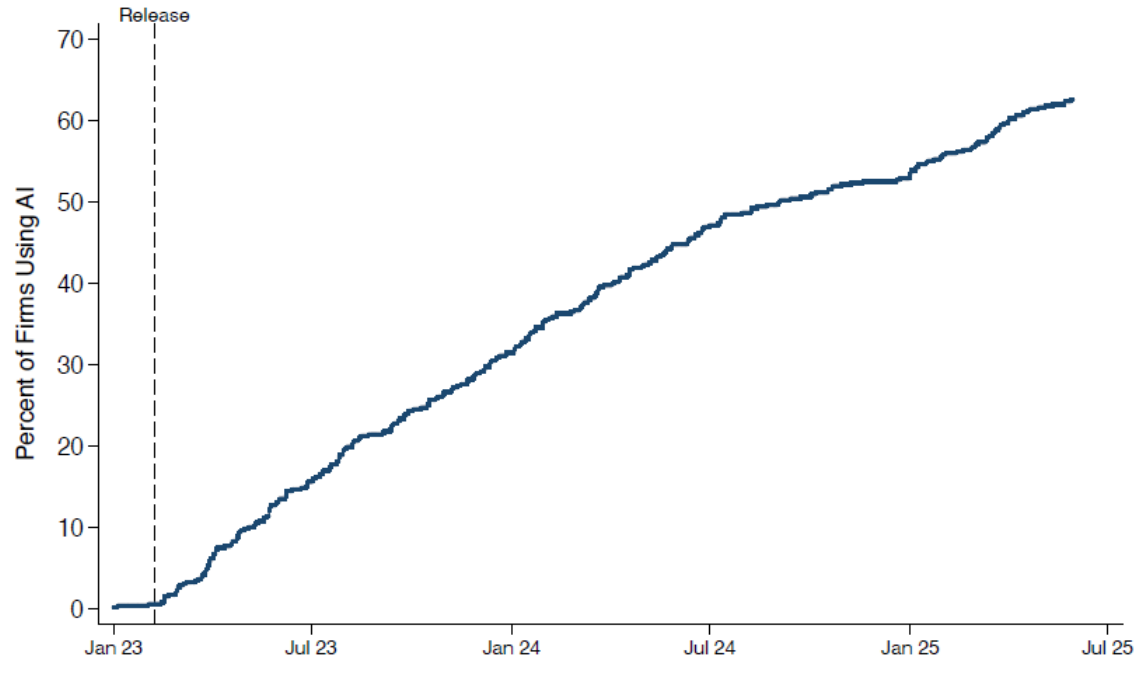
Challenge #2: Bottlenecks

Data from Copilot and Cursor deployments in the field

(Chen and Stratton 2026 - 100k engineers in 500 firms)

Firm and worker adoption

(Chen & Stratton, 2026)

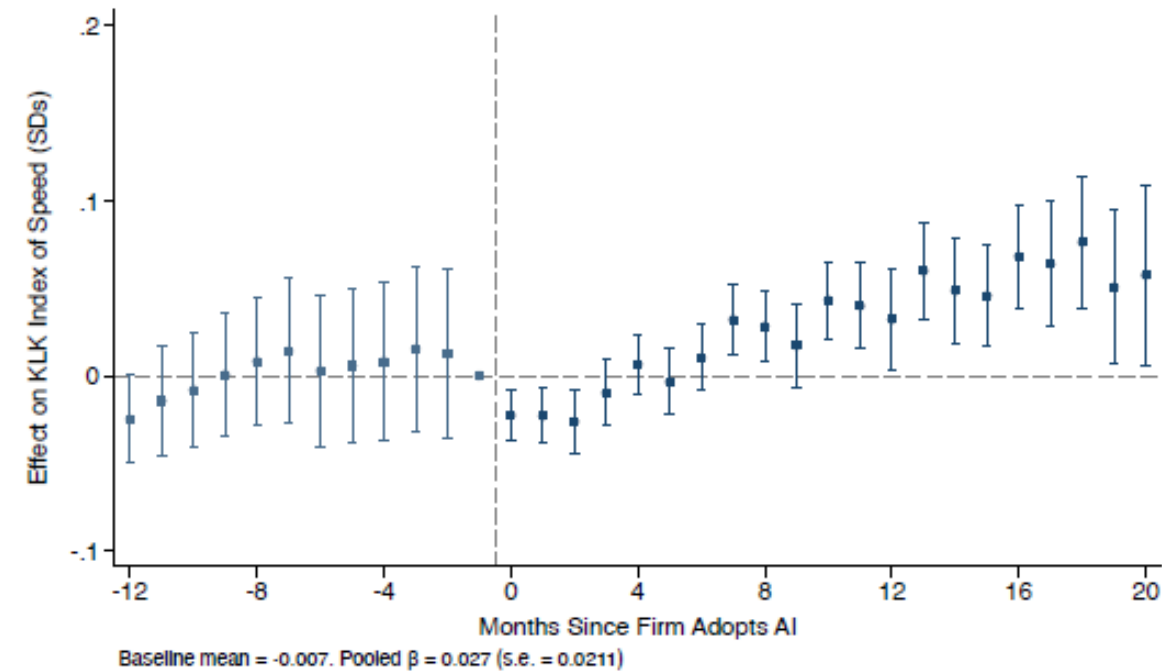


Note: firms NOT representative, based on use of Jellyfish productivity tool

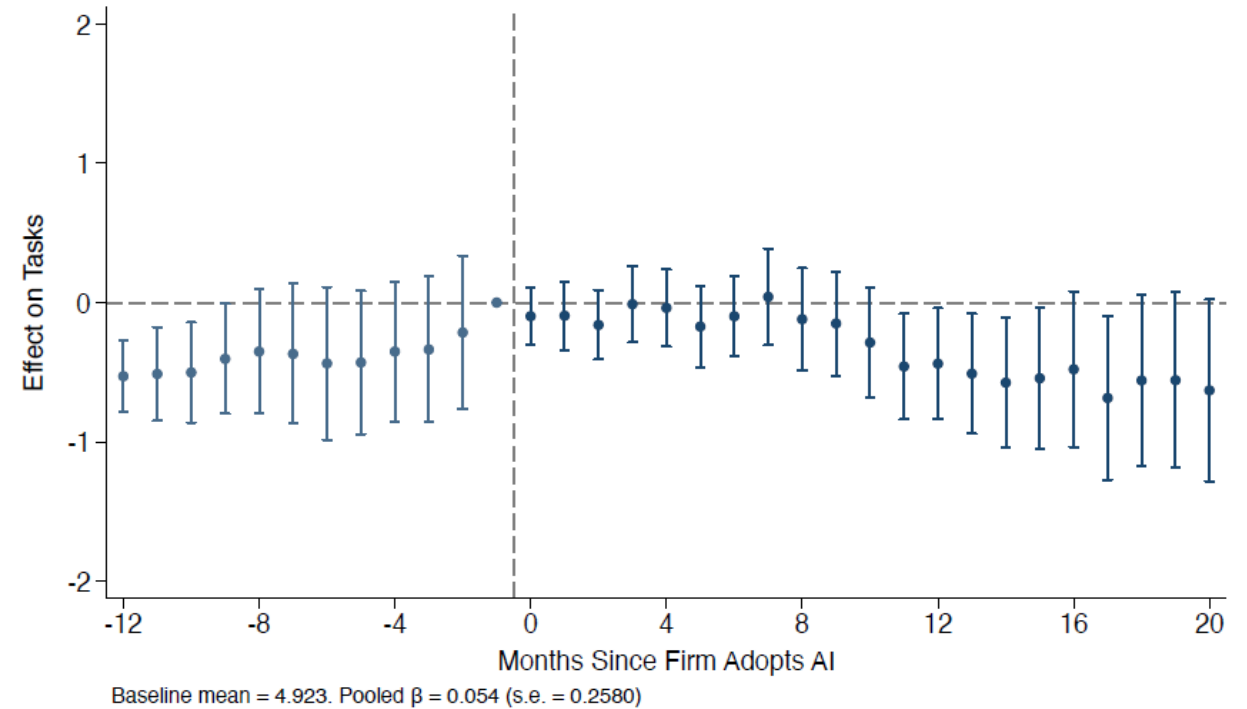
Effect on productivity

(Chen & Stratton, 2026)

A. Index of completion speed



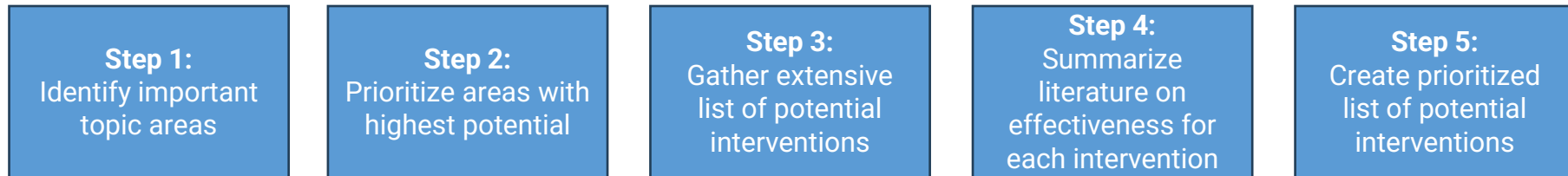
A. Number of tasks



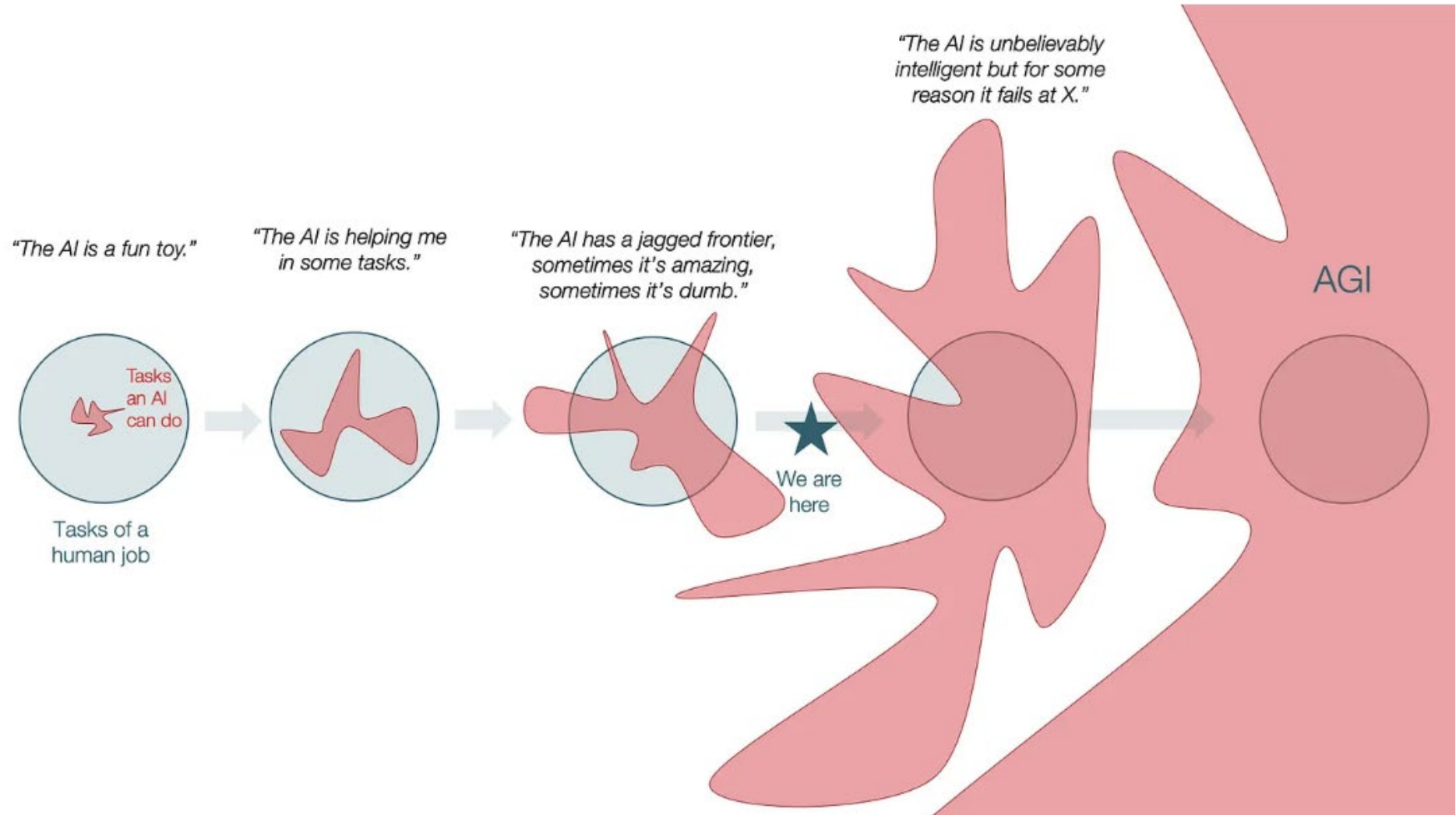
Note: no change in code quality, task size; employment. Similar across seniority

Motivating example: Automating opportunity identification in philanthropy

Agentic Flow



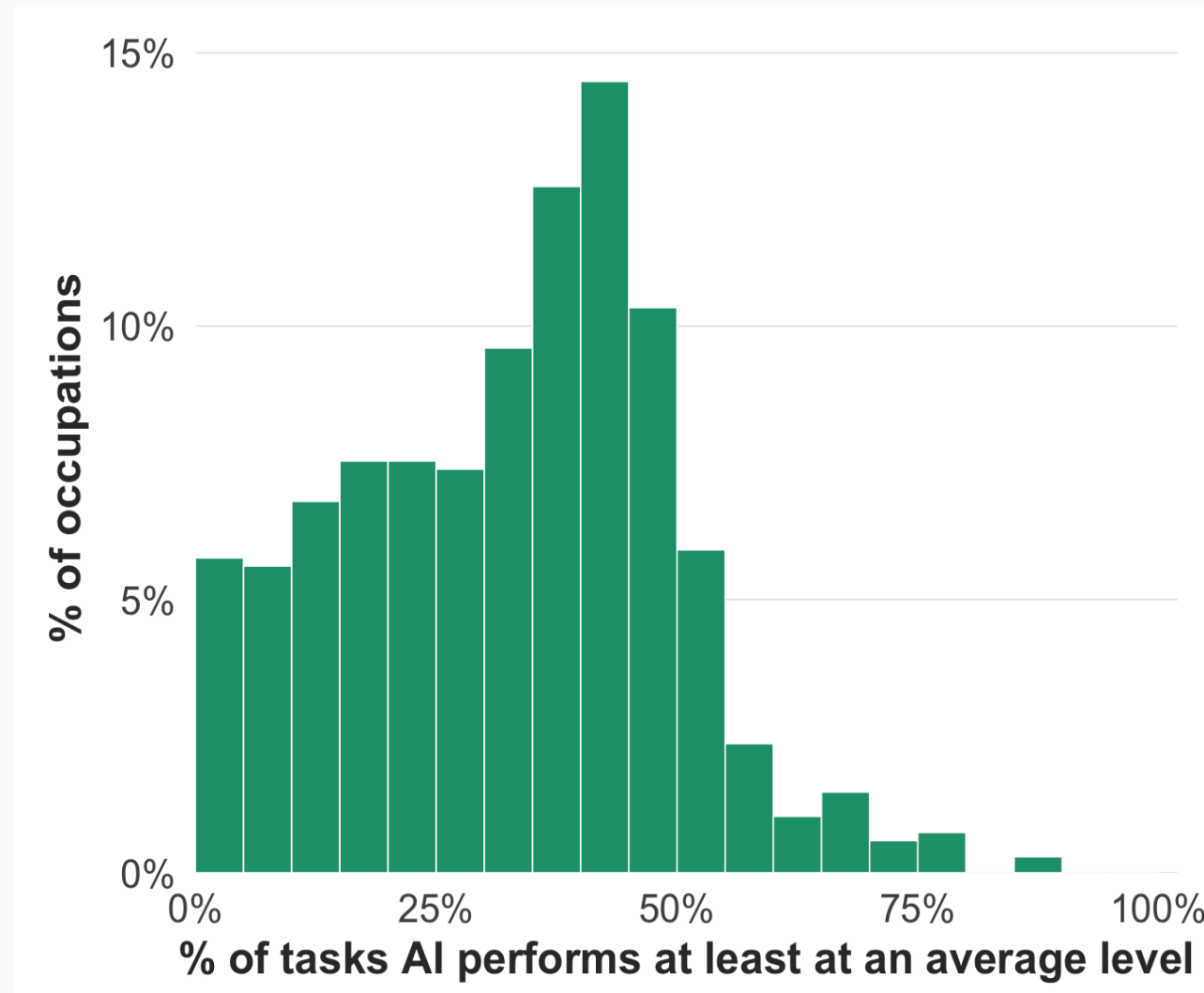
Jagged Frontier



Source: Mollick (2025)

Virtually no occupations can be fully automated by AI

(LLM-addressable tasks only shown)



Implication: biggest gains will come from re-designing processes around AI



Why was the Ford Model T only available in black?

Takeaways

- AI capabilities are impressive and growing quickly
- AI improvements have been driven by:
 - Scaling up computation *much faster than hardware progress* (Most important)
 - Improving algorithms (Both overall and with company secret sauce)
- Means that cutting-edge models are getting much more expensive
 - While previously achieved levels of performance are becoming rapidly less expensive
- So far, AI adoption and ROI have been disappointing
- Many reasons for AI failures
 - Benefits not large enough to offset business risks
 - Bottlenecks and jagged AI capabilities
- Biggest gains: redesign processes to focus on AI achievable tasks

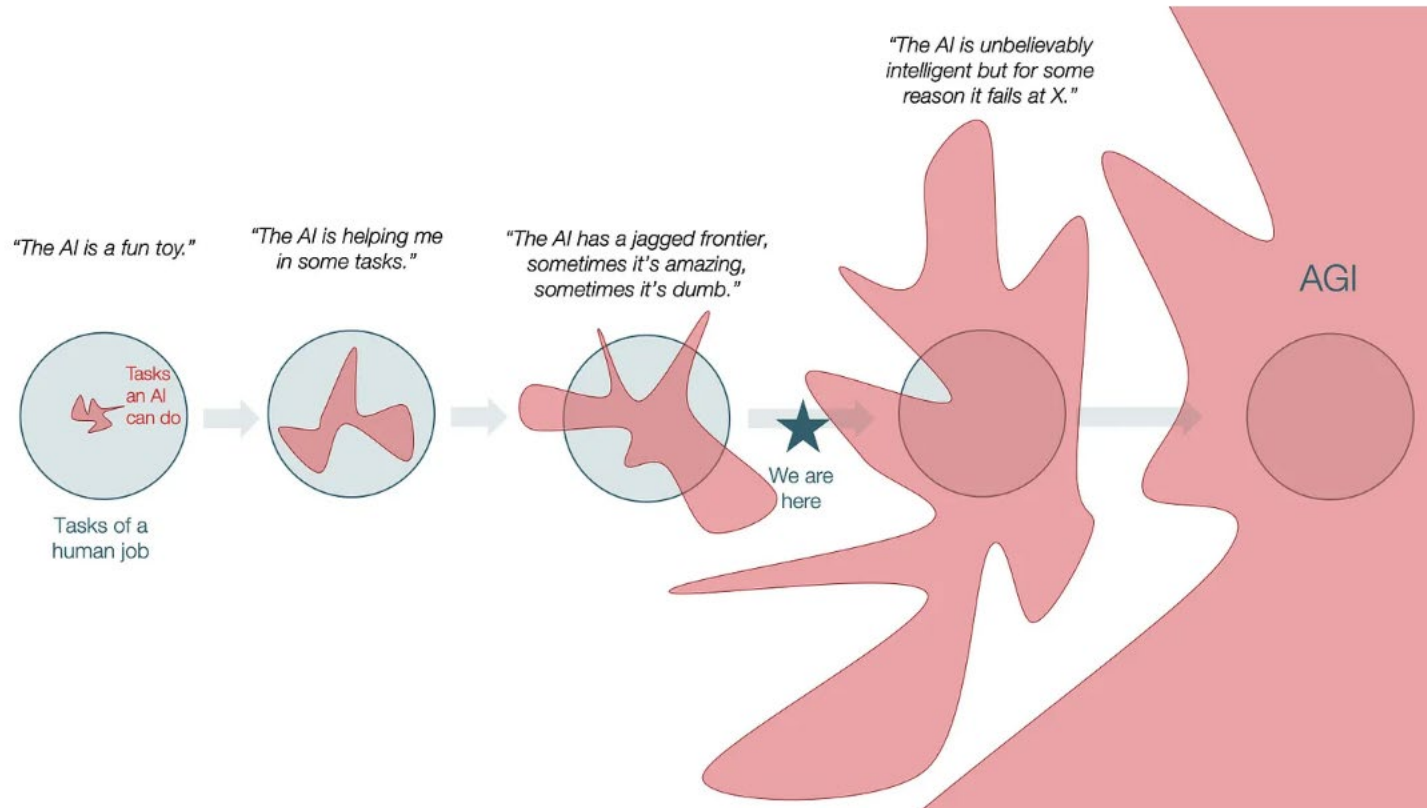
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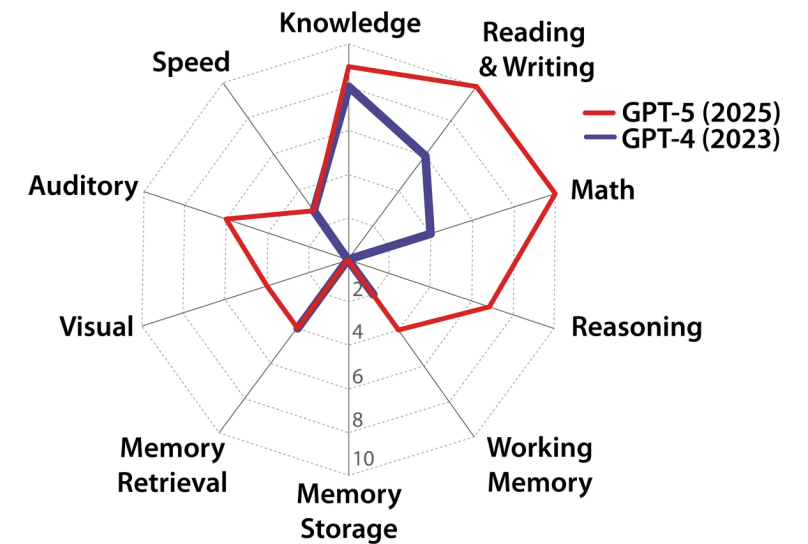


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Jagged Frontier



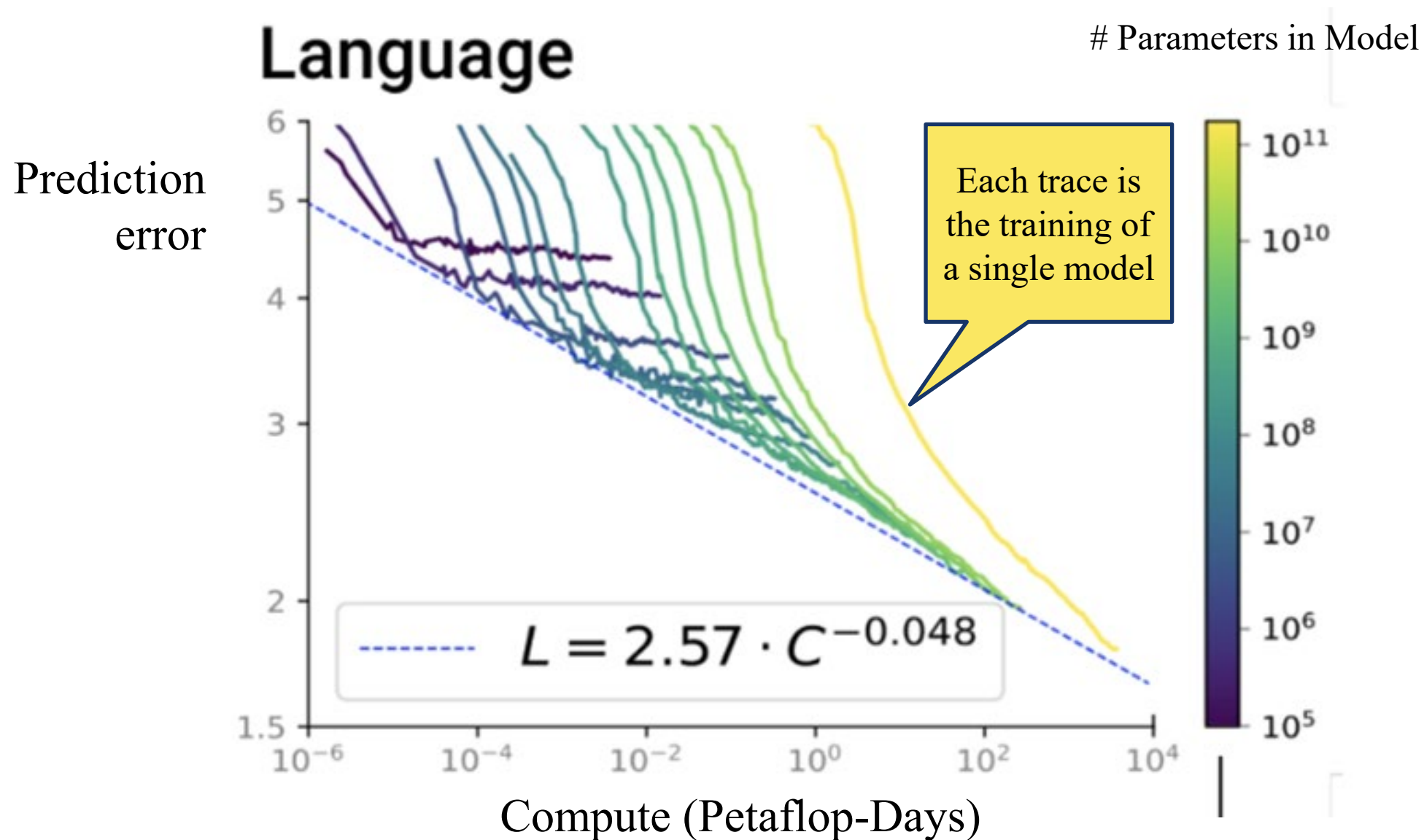
Source: Mollick (2025)



Source: Hendrycks et al (2025)

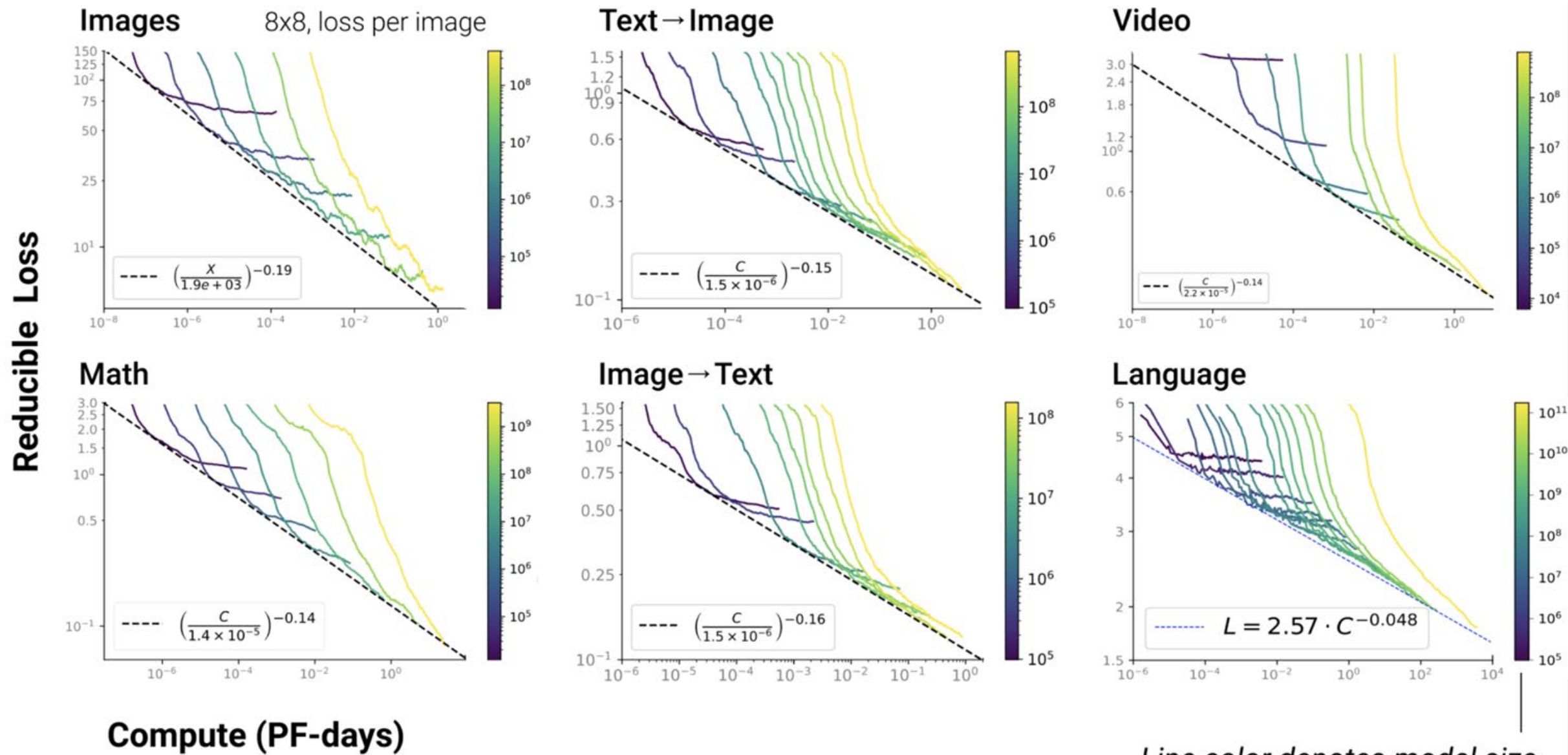
Emergence of scaling laws

(Henighan et al, 2020)



True across fields

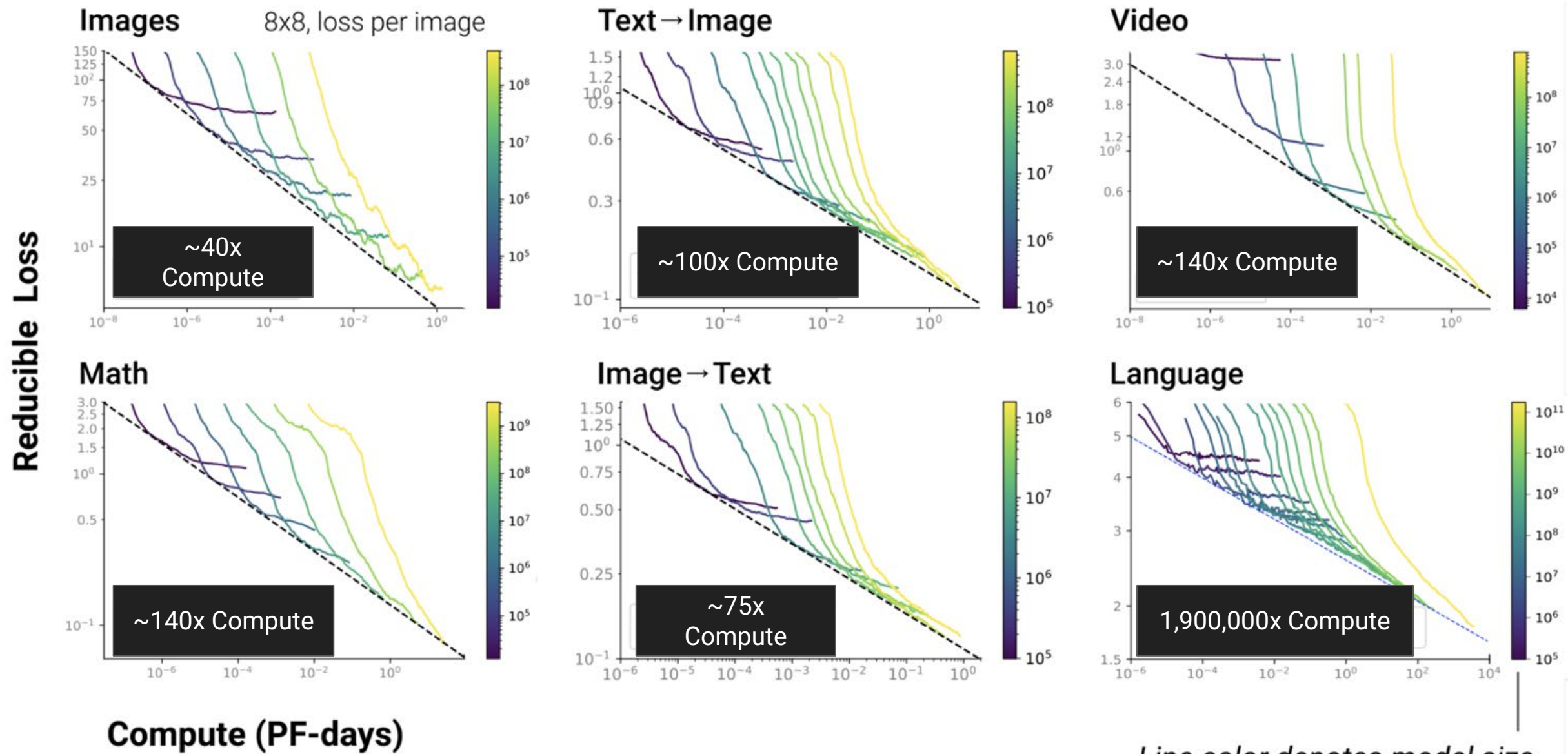
(Henighan et al, 2020)



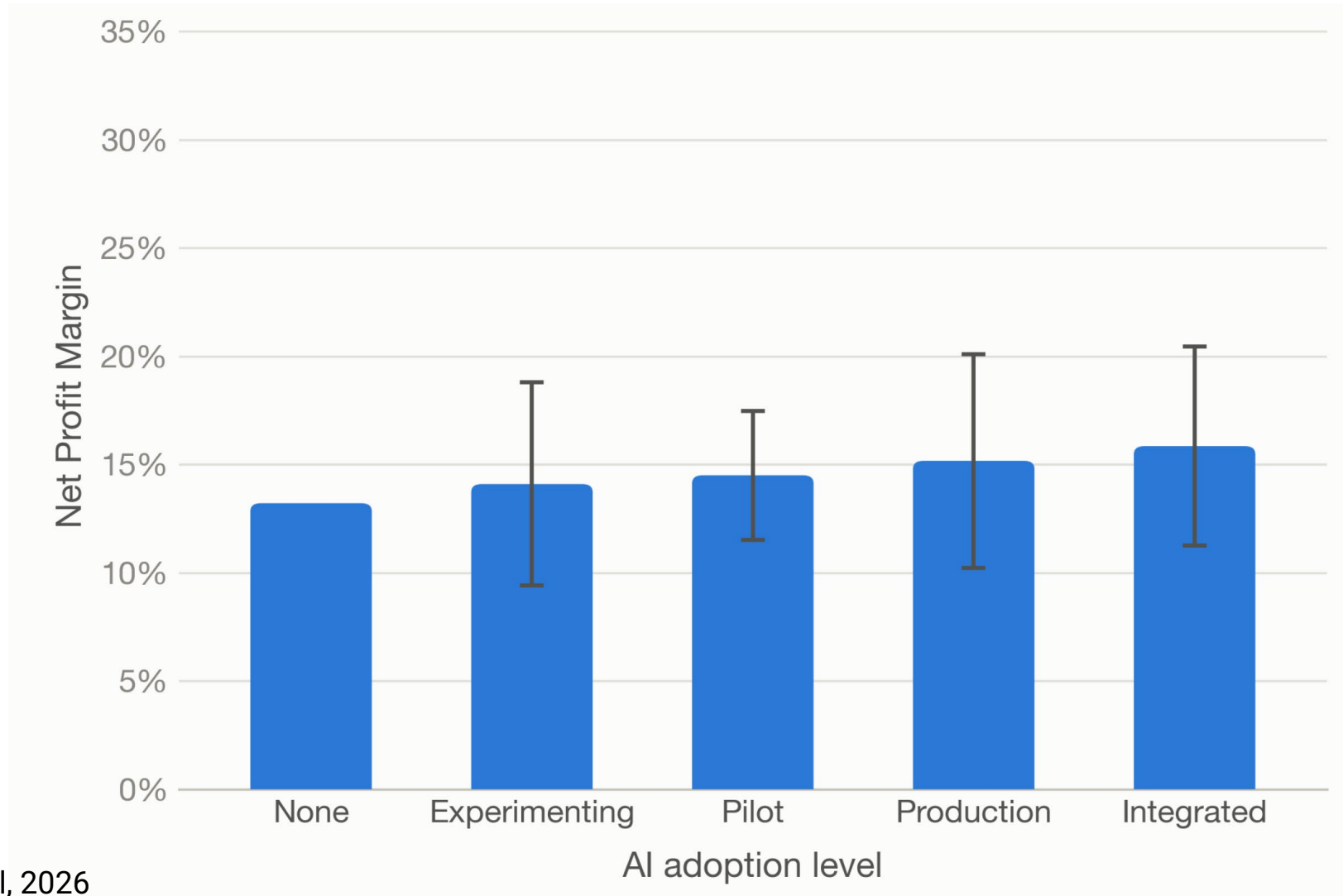
True across fields

(Henighan et al, 2020)

$\frac{1}{2}$ Loss $\rightarrow$



Tech firms that adoption AI more have higher profit margins



AliExpress: Mass Customization

(Fang et al., 2026)



Marketing Message

- Generate millions of unique messages targeted to individual consumers
- Effect: **+1.6% sales increase**
- *Note: limited deployment*

Risk: *Exposure to toxic content*

Risk description

AI that exposes users to harmful, abusive, unsafe or inappropriate content. May involve providing advice or encouraging action.

Real-world incidents

- YouTube Kids displayed violent cartoons and conspiracy theories to children (# 1)
- Google Gemini sent threatening "Please die" message to student (# 845)
- LA Times AI defended KKK in response to anti-hate article (# 946)

Discrimination &
Toxicity

Toxic content



MIT AI Risk
Repository

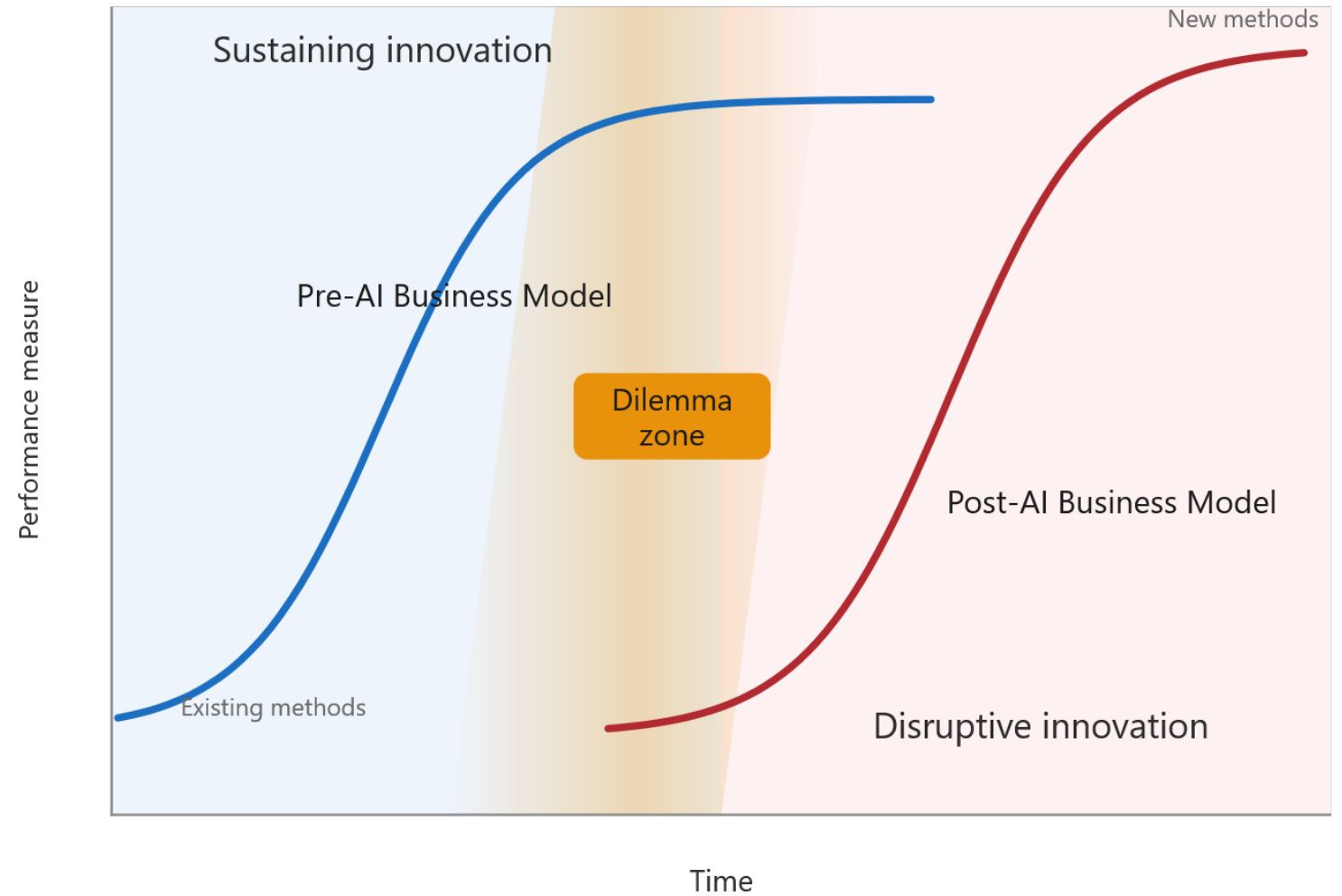
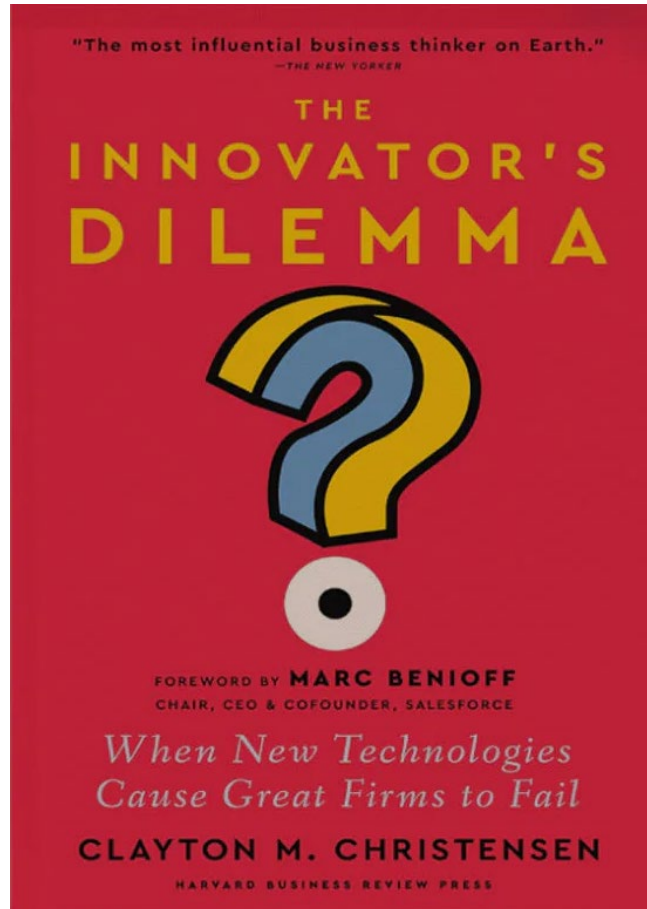
airisk.mit.edu



incidentdatabase.ai

There are many AI risks

Discrimination & Toxicity	Discrimination	Toxic content	Unequal performance
Privacy & Security	Loss of privacy	AI security vulnerabilities	
Misinformation	False information	Loss of consensus reality	
Malicious Actors & Misuse	Disinformation & influence	AI weapons & cyberattacks	AI fraud & scams
Human-Computer Interaction	Overreliance & unsafe use	Loss of human agency	
Socioeconomic & Environmental Harm	Power Centralization	Inequality & unemployment	Devaluation of human creativity
	Governance failure	Competitive dynamics	Environmental harm
AI System Safety, Failures, & Limitations	AI misalignment	Capability & robustness	AI welfare
	Dangerous capabilities	Transparency & interpretability	Multi-agent risks



Filling “long-tail” needs

Respond to ill-served customers...

- 24 hours per day
- In their own language
- Where previously no coverage (e.g. Pre-service questions)



AliExpress: Example of long-tail needs (Fang et al., 2026)

Pre-purchase service

- Pre-service chatbot
 - Previously: none
- 24/7, all languages
- Effect: **Sales +16.3%**

Translating product searches more accurately

- Use AI to better translate context of query
- Improves platform search engine's performance
- Effect: **Sales +2.9%**

Agents and Long-tasks

Example: Accuracy in multiplying numbers

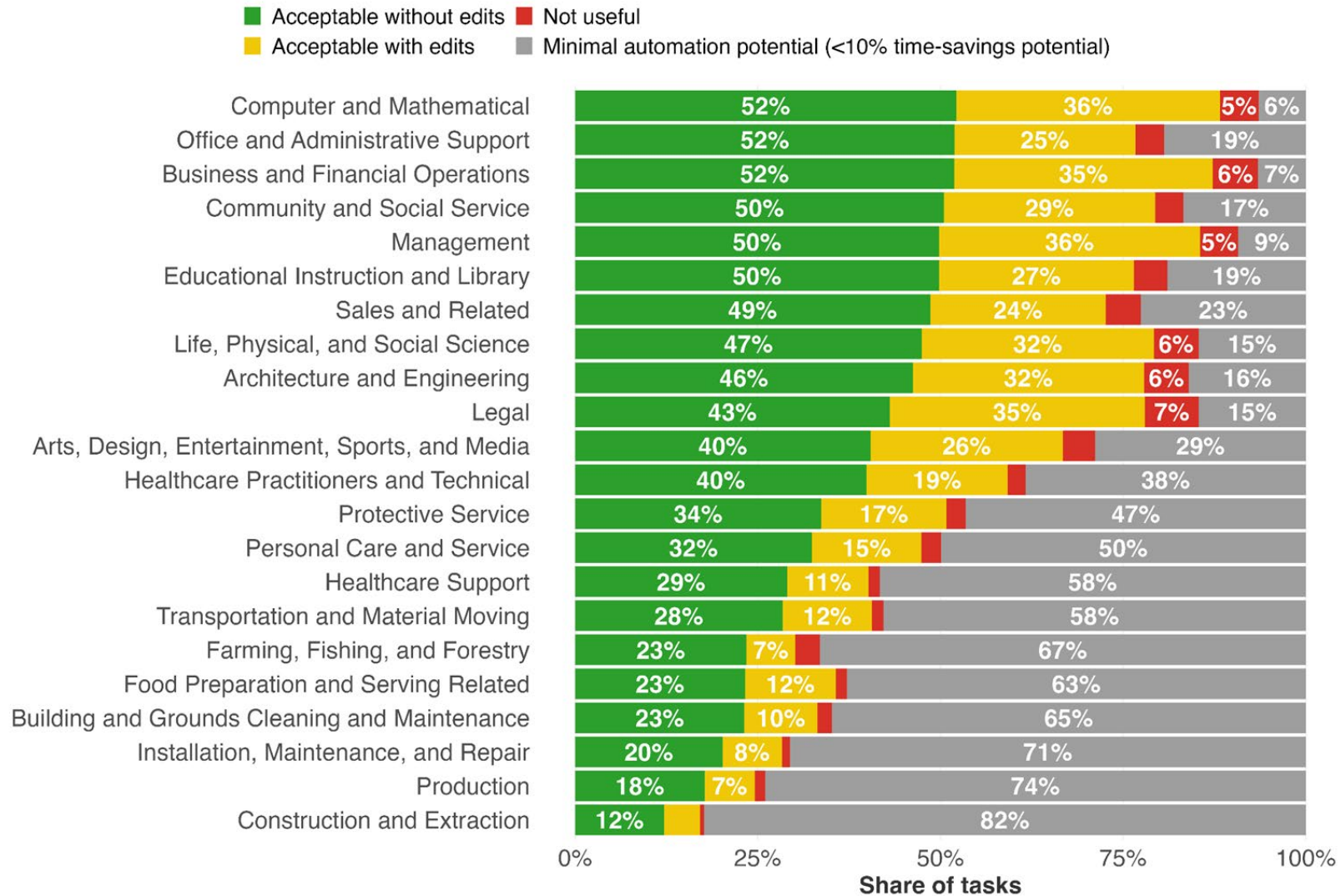


Accuracy of o1-mini
Digits in Number 1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	100	100	100	100	96.2	100	96.2	100	96.2	100	92.3	100	88.5	92.3	96.2	88.5	92.3	80.8	92.3	88.5
2	100	100	100	100	100	100	100	96.2	100	92.3	100	84.6	69.2	76.9	80.8	69.2	65.4	80.8	80.8	46.2
3	100	100	100	92.3	96.2	92.3	100	100	88.5	84.6	76.9	84.6	73.1	57.7	57.7	65.4	53.8	34.6	42.3	26.9
4	100	100	100	100	100	100	92.3	92.3	88.5	92.3	84.6	73.1	53.8	42.3	50.0	46.2	46.2	30.8	11.5	26.9
5	100	100	100	92.3	96.2	92.3	88.5	76.9	76.9	69.2	57.7	38.5	65.4	61.5	34.6	23.1	26.9	30.8	7.7	3.8
6	96.2	96.2	92.3	100	92.3	84.6	69.2	73.1	61.5	57.7	61.5	46.2	19.2	15.4	15.4	23.1	11.5	0.0	15.4	7.7
7	96.2	100	92.3	100	80.8	76.9	61.5	73.1	50.0	57.7	46.2	46.2	26.9	11.5	11.5	7.7	3.8	11.5	3.8	7.7
8	96.2	100	88.5	92.3	84.6	69.2	65.4	61.5	57.7	61.5	34.6	26.9	7.7	23.1	3.8	0.0	3.8	0.0	3.8	0.0
9	100	100	100	80.8	57.7	57.7	50.0	50.0	53.8	19.2	34.6	19.2	3.8	3.8	3.8	0.0	15.4	0.0	0.0	0.0
10	96.2	96.2	96.2	80.8	73.1	50.0	30.8	34.6	19.2	3.8	0.0	7.7	0.0	3.8	0.0	0.0	0.0	0.0	0.0	0.0
11	96.2	96.2	84.6	73.1	57.7	42.3	23.1	26.9	11.5	3.8	7.7	7.7	3.8	3.8	0.0	0.0	0.0	0.0	0.0	0.0
12	92.3	92.3	69.2	69.2	50.0	23.1	3.8	7.7	15.4	11.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
13	88.5	80.8	84.6	57.7	42.3	26.9	7.7	0.0	3.8	7.7	3.8	0.0	3.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
14	84.6	76.9	73.1	65.4	38.5	11.5	3.8	3.8	3.8	0.0	0.0	0.0	3.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
15	96.2	84.6	73.1	34.6	7.7	26.9	3.8	3.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
16	88.5	76.9	57.7	23.1	19.2	7.7	7.7	3.8	3.8	0.0	0.0	0.0	0.0	0.0	3.8	0.0	0.0	0.0	0.0	0.0
17	84.6	88.5	65.4	38.5	23.1	11.5	11.5	3.8	3.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
18	92.3	69.2	53.8	30.8	11.5	19.2	3.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
19	80.8	61.5	34.6	15.4	3.8	3.8	3.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
20	84.6	61.5	26.9	23.1	3.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Digits in Number 2

How good are frontier models?



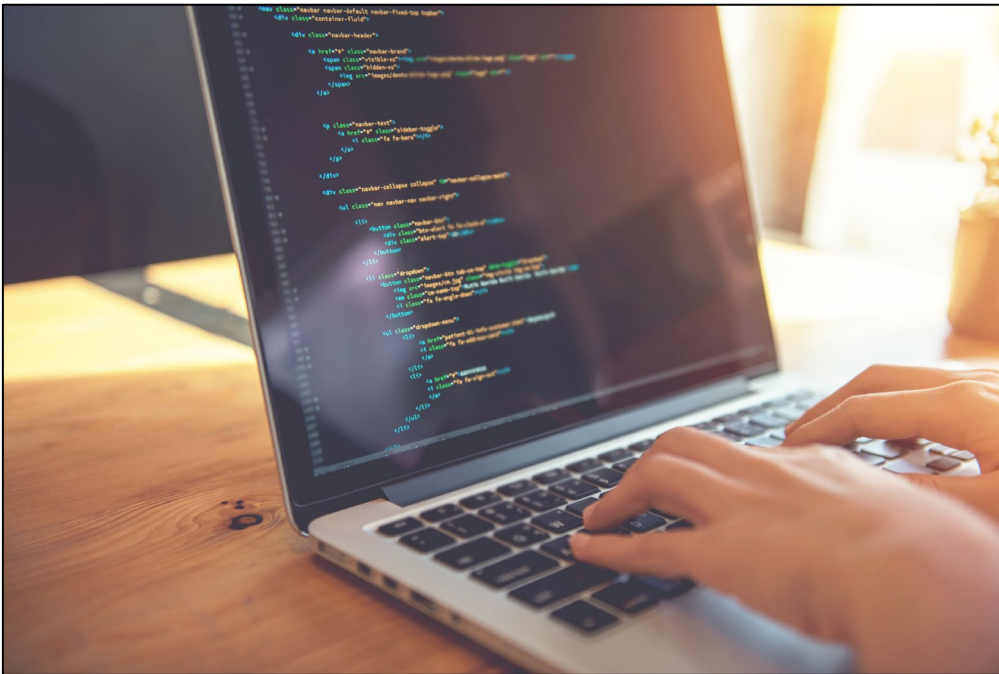
**Implications for when someone says:
“Let’s automate this with an agent”**

This is a general problem

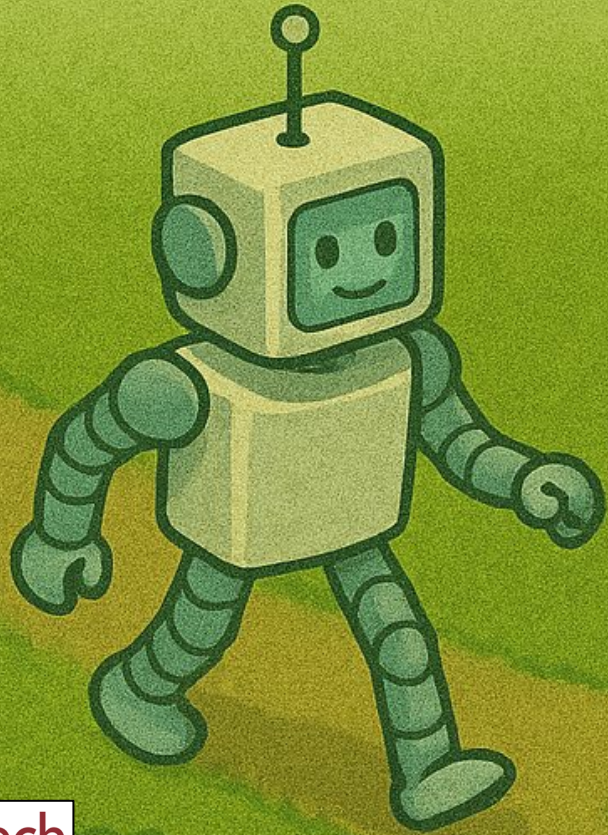


**Could still produce cost savings if AI cheaper, but
larger step change can't happen**

Bottlenecks in Practice: Coding → Code Review



AI'S GARDEN OF BRANCHING PATHS



What happens if solving the problem requires multiple steps?

(Insight due to Yann LeCun)

1 st Step	2 nd Step	3 rd Step	All Steps
95%			95%
95%	95%		90%
95%	95%	95%	86%

Agentic Example: Automated Vending Machine



Anthropic starts testing customer-facing AI agents with financial authority